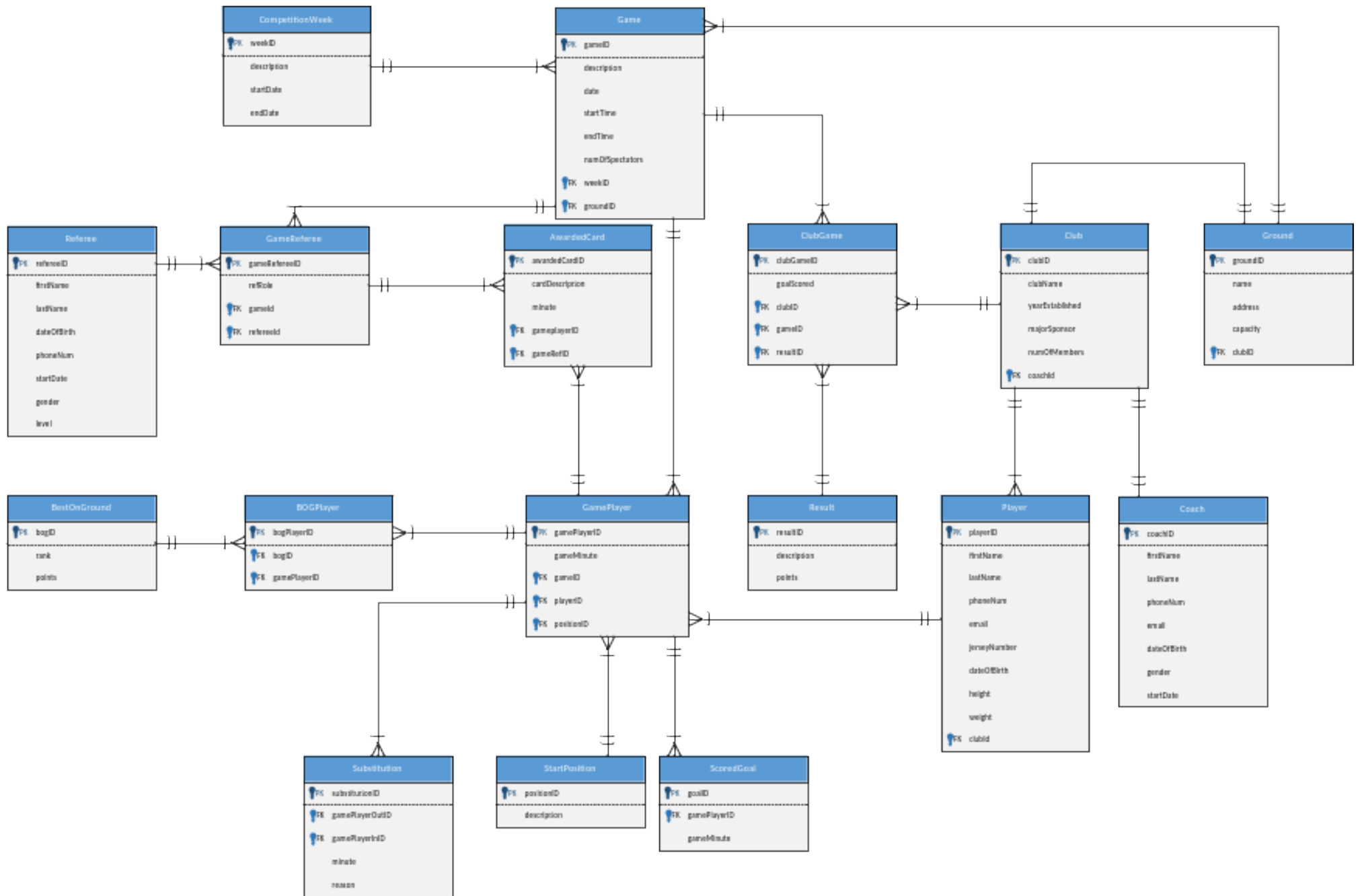

1. Design / Conceptual DB

1.1 Entity Relationship Diagram



1.2 Business Rules

1. CompetitionWeek to Game

- 1a. A CompetitionWeek includes one or more than one Game.
- 1b. A Game is played in one and only one CompetitionWeek.

2. Game and GameReferee

- 2a. A Game appears in one or more than one GameReferee.
- 2b. A GameReferee supervises one and only one Game.

3. Referee and GameReferee

- 3a. A Referee appears in one or more than one GameReferee
- 3b. A GameReferee records one and only one Referee.

4. Ground and Game

- 4a. A Ground hosts one or more than one Game.
- 4b. A Game takes place at one and only one Ground.

5. Game and ClubGame

- 5a. A game is recorded in one or more than one ClubGame.
- 5b. A ClubGame records one and only one Game.

6. Club and ClubGame

- 6a. A ClubGame records one and only one Club.
- 6b. A Club plays in one or more than one ClubGame.

7. Club and Ground

- 7a. A Club plays in one and only one Ground.
- 7b. A Ground hosts one and only one Club.

8. Game and Gameplayer

- 8a. A GamePlayer contains one and only one game
- 8b. A Game is recorded in one or more than one GamePlayer.

9. GamePlayer and AwardedCard

- 9a. A GamePlayer gets one or more than one AwardedCard.
- 9b. An AwardedCard records one and only one GamePlayer.

10. GameReferee to AwardedCard

- 10a. A GameReferee appears in one or more than one AwardedCard.
- 10b. An AwardedCard is issued by one and only one GameReferee.

11. BestOnGround and BOGPlayer

- 11a. A BestOnGround appears in one or more than one BOGPlayer.
- 11b. A BOGPlayer records one and only one BestOnGround.

12. GamePlayer and BOGPlayer

- 12a. A GamePlayer appears in one or more than one BOGPlayer.
- 12b. A BOGPlayer records one and only one GamePlayer.

13. Result and ClubGame

- 13a. A Result records one or more than one ClubGame.
- 13b. A ClubGame has one and only one Result.

14. Club and Player

14a. A Club has one or more than one Player.

14b. A Player participates in one and only one club.

15. Club and Coach

15a. A Club has one and only one Coach.

15b. A Coach works for one and only one Club.

16. Player and GamePlayer

16a. A Player participates in one or more than one GamePlayer.

16b. A GamePlayer records one and only one Player.

17. GamePlayer and ScoredGoal

17a. A GamePlayer gets one or more than one ScoredGoals.

17b. A ScoredGoal is recorded in one and only one GamePlayer.

18. StartPosition and GamePlayer

18a. A StartPosition is assigned to one or more than one GamePlayers.

18b. A GamePlayer has one and only one StartPosition.

19. GamePlayer and Substitution

19a. A GamePlayer participates in one or more than one Substitution.

19b. A Substitution appears in one and only one GamePlayer.

2. Schema / Logical DB Schema

2.1 DDL-Schema

TABLE	COLUMNS

-	
COACH	(<u>coachID</u> [PK], firstName, lastName, phoneNum, email, dateOfBirth, gender, startDate)
COMPETITIONWEEK	(<u>weekID</u> [PK], description, startDate, endDate)
REFEREE	(<u>refereeID</u> [PK], firstName, lastName, dateOfBirth, phoneNum, starDate, gender, refLevel)
STARTPOSITION	(<u>positionID</u> [PK], description)
BESTONGROUND	(<u>bogID</u> [PK], rank, points)
RESULT	(<u>resultID</u> [PK], description, points)
CLUB	(<u>clubID</u> [PK], clubName, yearEstablished, majorSponsor, numOfMembers, <i>coachID</i> [FK])
PLAYER	(<u>playerID</u> [PK], firstName, lastName, phoneNum, email, jerseyNumber, dateOfBirth, height, weight, <i>clubID</i> [FK])
GROUND	(<u>groundID</u> [PK], name, address, capacity, <i>clubID</i> [FK])
GAME	(<u>gameID</u> [PK], description, gameDate, startTime, endTime, numOfSpectators, <i>weekID</i> [FK], <i>groundID</i> [FK])
CLUBGAME	(<u>clubGameID</u> [PK], <i>gameID</i> [FK], <i>clubID</i> [FK], goalScored, <i>resultID</i> [FK])
GAMEREFEREE	(<u>gameRefereeID</u> [PK], refRole, <i>gameID</i> [FK], <i>refereeID</i> [FK])
GAMEPLAYER	(<u>gamePlayerID</u> [PK], <i>gameID</i> [FK], <i>playerID</i> [FK], <i>positionID</i> [FK], gameMinute)
BOGPLAYER	(<u>bogPlayerID</u> [PK], <i>bogID</i> [FK], <i>gamePlayerID</i> [FK])
SUBSTITUTION	(<u>substitutionID</u> [PK], <i>gamePlayerOutID</i> [FK], <i>gamePlayerInID</i> [FK], minute, reason)
AWARDEDCARD	(<u>awardedCardID</u> [PK], cardDescription, minute, <i>gamePlayerID</i> [FK], <i>gameRefID</i> [FK])
SCOREDGOAL	(<u>goalID</u> [PK], <i>gamePlayerID</i> [FK], gameMinute)

-	

KEY: = PK, = FK

2.2 Data Dictionary

Format details: FONT=Calibri, size=8, color=black. PARAGRAPH space before=0, space after=0. TABLE

 bg-color=RGB/204,102,0

 border-color=RGB/153,51,0.

TableName	Alias	Column Name	Description	Format	Opt/Require	PK Name	PK Column Ref	FK Name	FK Column Ref	Table Ref	Other C.Name(S)
Coach	ch	coachID	Coach's unique ID and PK in Coach	CHAR (4)	NOT NULL	Coach_PK	coachID				
		firstName	Coach's first name	VARCHAR2 (25)	NOT NULL						
		lastName	Coash's last name	VARCHAR2 (25)	NOT NULL						
		phoneNum	Coach's phone number	CHAR (10)							
		email	Coach's email address	VARCHAR2(30)							
		dateOfBirth	Date of birth of a coach	DATE							
		gender	Coach's gender	CHAR (1)							
		startDate	The coach's employment start date	DATE							
Competition Week	cw	weekID	Unique ID of competition week and PK in CompetitionWeek	CHAR (4)	NOT NULL	CompetitionWeek_PK	weekID				
		description	Description of competition week	VARCHAR2(25)	NOT NULL						
		startDate	Start date of competition week	DATE	NOT NULL						
		endDate	End date of competition week	DATE	NOT NULL						
Referee	ref	refereeID	Unique identification of referee and PK in Referee	CHAR (4)	NOT NULL	Referee_PK	refereeID				
		firstName	Referee's first name	VARCHAR2(25)	NOT NULL						
		lastName	Referee's last name	VARCHAR2(25)	NOT NULL						
		dateOfBirth	Referee's date of birth	DATE							
		phoneNum	Referee's phone number	CHAR(10)							
		startDate	Referee's employment start date	DATE							
		gender	The gender of the referee	CHAR(1)							

			(M/W)								
		refLevel	The qualification of the referee (1-4)	CHAR (1)							
StartPosition	sp	positionID	Unique ID of a pitch start position and PK in StartPosition	CHAR(4)	NOT NULL	StartPosition_PK	positionID				
		description	Name of the pitch start position	VARCHAR(40)	NOT NULL						
BestOnGround	bog	bogID	Unique ID and PK in BestOnGround. A BestOnGround record holds information about performance scoring data such as rank and associated points.	CHAR(3)	NOT NULL	BestOnGround_PK	bogID				
		rank	Rank associated with the record	CHAR(1)							
		Points	Points associated with the record	NUMBER(1,0)							
Result	rs	resultID	The unique ID of a result	CHAR(4)	NOT NULL	Result_PK	resultID				
		description	The type of result (HomeWin, AwayWin, HomeDraw, AwayDraw, HomeLoss, AwayLoss)	VARCHAR(15)	NOT NULL						
		points	Points associated with a result	NUMBER(1,0)	NOT NULL						
Club	cl	clubID	Unique ID of a club and PK in CLub	CHAR (4)	NOT NULL	Club_PK	clubID				
		clubName	CLub's name	VARCHAR 2 (25)	NOT NULL						
		yearEstablished	Club's establishment year	NUMBER (4,0)							
		majorSponsor	Club's major sponsor	VARCHAR2 (25)							
		numOfMembers	Club's total member	NUMBER (5,0)							
		coachID	CLubs's coach and FK from Coach	CHAR (4)	NOT NULL			Coach_FK	CoachID	Coach	

Player	pl	playerID	Unique ID of a PLayer	CHAR (5)	NOT NULL	Player_PK	playerID				
		firstName	Player's first name	VARCHAR2 (30)	NOT NULL						
		lastName	Player's last name	VARCHAR2 (30)	NOT NULL						
		phoneNum	Player's phone number	CHAR (10)							
		email	Player's email address	VARCHAR2 (30)							
		jerseyNumber	Jersey/shirt number	VARCHAR2 (2)							
		dateOfBirth	Player's date of birth	DATE							
		height	Player's height (cm)	NUMBER(3,0)							
		weight	Player's weight (kg)	NUMBER (3,0)							
		clubID	The club a player plays for and FK from Club	CHAR (4)	NOT NULL			Player_Club_FK	clubID	Club	
Ground	grn	groundID	Unique ID of a ground	CHAR (4)	NOT NULL	Ground_PK	groundID				
		name	Ground's name	VARCHAR2 (30)	NOT NULL						
		address	Ground's address	VARCHAR 2 (40)	NOT NULL						
		capacity	Ground capacity	NUMBER (5,0)	NOT NULL						
		clubID	Club who trains on the ground and FK from Club	VARCHAR (4)	NOT NULL			Ground_Club_FK	clubID	Club	
Game	gm	gameID	The unique ID of a game	CHAR (4)	NOT NULL	Game_PK	gameID				
		description	Game description (fixture)	VARCHAR (15)	NOT NULL						
		gameDate	Date a game takes place on	DATE	NOT NULL						
		startTime	The start time of a game.	TIME	NOT NULL						
		endTime	The end time of the game.	TIME	NOT NULL						
		numOfSpectators	Number of people watching the game	NUMBER (5,0)							
		weekID	Week which a game plays in and	CHAR (4)	NOT NULL			Game_	weekID	CompetitionWeek	

			FK from Week					CompetitionWeek_FK			
		groundID	Ground a game play on.	CHAR (4)	NOT NULL			Game_ Ground_FK	groundID	Ground	
ClubGame	cg	clubGameID	The unique ID of a record kept about a game and its participating club. It is also a PK in ClubGame	CHAR (5)	NOT NULL	clubGame_PK	clubGameID				
		gameID	ID of a game and FK from Game	CHAR (4)	NOT NULL			clubGame_Game_FK	gameID	Game	
		clubID	ID of a club and FK from Club	CHAR (4)	NOT NULL			clubGame_Club_FK	clubID	Club	
		goalScored	The number of goals scored in the game	NUMBER (2,0)	NOT NULL						
		resultID	Result ID and FK from Result	CHAR (4)	NOT NULL			clubGame_Result_FK	resultID	Result	
GameReferee	gr	gameRefereeID	Unique ID of a record kept about a game its associated Referee	CHAR (6)	NOT NULL	GameReferee_PK	GameRefereeID				
		refRole	The role of referee in the game	VARCHAR2 (15)	NOT NULL						
		gameID	The game which the referee refereed in and FK from Game	CHAR (4)	NOT NULL			GR_Game_FK	gameID	Game	
		refereeID	The referee associated with the game and FK from Referee	CHAR (5)	NOT NULL			GR_Referee_FK	refereeID	Referee	
GamePlayer	gp	gamePlayerID	Unique ID of a record of each player's participation in a game and PK in GamePlayer	CHAR(6)	NOT NULL	GamePlayer_PK	gamePlayerID				
		gameID	The game the player played in and FK from Game	CHAR(5)	NOT NULL			GamePlayer_ Game_FK	gameID	Game	
		playerID	The player who played in the game and FK from Player	CHAR(5)	NOT NULL			GamePlayer_ Player_FK	playerID	Player	
		positionID	Starting position of the player and FK from StartPosition	CHAR(5)	NOT NULL			GamePlayer_ Position_FK	positionID	StartPosition	

		gameMinute	Minutes the player played for in the game	NUMBER (3,0)	NOT NULL						
Substitution	sub	substitutionID	Unique ID of each substitution made in a game and PK in Substitution	CHAR (6)	NOT NULL	Substitution_PK	substitutionID				
		gamePlayerOutID	The ID of a GamePlayer record where its associated player got substituted out and FK from GamePlayer	CHAR (6)	NOT NULL			GamePlayerOut_ID_FK	gamePlayerID	GamePlayer	
		gamePlayerInID	The ID of a GamePlayer record where its associated player got substituted in and FK from GamePlayer	CHAR (6)	NOT NULL			GamePlayerIn_ID_FK	gamePlayerID	GamePlayer	
		minute	Minute the substitution happened at	NUMBER (3,0)	NOT NULL						
		reason	The reason for substitution	VARCHAR2 (30)							
AwardedCard	ac	awardedCardID	Unique ID of each awarded card PK in AwardedCard	CHAR(5)	NOT NULL	AwardedCard_PK	awardedCardID				
		cardDescription	Type of card (YellowCard, RedCard)	VARCHAR2 (6)	NOT NULL						
		minute	Minute of the game when the card was awarded	NUMBER(3,0)	NOT NULL						
		gamePlayerID	An ID of GamePlayer record where its player was awarded a card and FK from GamePlayer	CHAR (6)	NOT NULL			AwardedCard_ GamePlayer_FK	gamePlayerID	GamePlayer	
		gameRefereeID	An ID of a record of GameReferee associated referee in the game awarded the card.	CHAR (6)	NOT NULL			AwardedCard_ GameReferee_FK	gameRefereeID	GameReferee	
ScoredGoal	sg	goalID	A unique ID of a goal and PK in ScoredGoal	CHAR (5)	NOT NULL	ScoredGoal_PK	scoredGoalID				
		gamePlayerID	An ID of a record of GamePlayer	CHAR (6)	NOT NULL			ScoredGoal_	gamePlayerID	GamePlayer	

			that links a game and the player who scored the goal.					GamePlayer_FK			
		gameMinute	The minute at which the goal was scored	NUMBER (3,0)	NOT NULL						
BogPlayer	bp	bogPlayerID	Unique ID of a record kept identifying a player who was nominated a Best on Ground.	CHAR(6)	NOT NULL	BOGPlayer_PK	bogPlayerID				
		bogID	An ID of a record of BestOnGround and FK from BestOnGround.	CHAR(3)	NOT NULL			BogPlayer_ Bog_FK	bogID	BOG	
		gamePlayerID	An ID of a record of GamePlayer where its associated player got nominated as a best on ground and FK from GamePlayer	CHAR(6)	NOT NULL			BogPlayer_ GamePlayer_FK	bogID	GamePlayer	

COMPETITIONWEEK									
Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies
SQL	REST								
Create	Drop	Enable	Disable						
Constraint	Type	Search Condition	Related Constraint	Columns	Delete Rule	Status	Last Change	Index	Invalid
SYS_C00205319	Check	"WEEKID" IS NOT NULL	-	-	-	ENABLED	09/23/2025 07:05:43 AM	-	-
SYS_C00205320	Check	"DESCRIPTION" IS NOT NULL	-	-	-	ENABLED	09/23/2025 07:05:43 AM	-	-
SYS_C00205321	Check	"STARTDATE" IS NOT NULL	-	-	-	ENABLED	09/23/2025 07:05:43 AM	-	-
SYS_C00205322	Check	"ENDDATE" IS NOT NULL	-	-	-	ENABLED	09/23/2025 07:05:43 AM	-	-
COMPETITIONWEEK_PK	Primary	-	-	WEEKID	-	ENABLED	09/23/2025 07:05:43 AM	COMPETITIONWEEK_PK	-

Referee

The SQL Syntax:

```
CREATE TABLE Referee (
    refereeID CHAR (4) NOT NULL,
    firstName VARCHAR2(25) NOT NULL,
    lastName VARCHAR2(25) NOT NULL,
    dateOfBirth DATE,
    phoneNum CHAR(10),
    startDate DATE,
    gender CHAR(1),
    refLevel CHAR(1),
    CONSTRAINT Referee_PK PRIMARY KEY (refereeID)
);
```

Result of syntax in Oracle:

GAMEREFEREE

+ -

Table

Data

Indexes

Model

Constraints

Grants

Statistics

UI Defaults

Triggers

Dependencies

SQL

REST

Add Column

Modify Column

Rename Column

Drop Column

Rename

Copy

Drop

Truncate

Create Lookup Table

Column Name	Data Type	Nullable	Default	Primary Key
GAMEREFEREEID	CHAR(6)	No	-	1
REFROLE	VARCHAR2(15)	No	-	-
GAMEID	CHAR(4)	No	-	-
REFEREEID	CHAR(4)	No	-	-

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GAMEREFEREE											+ v
Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL	REST
Create	Drop	Enable	Disable								
Constraint	Type	Search Condition	Related Constraint			Columns	Delete Rule	Status	Last Change	Index	Invalid
SYS_C00205436	Check	"GAMEREFEREEID" IS NOT NULL	-			-	-	ENABLED	09/23/2025 09:01:06 AM	-	-
SYS_C00205437	Check	"REFROLE" IS NOT NULL	-			-	-	ENABLED	09/23/2025 09:01:06 AM	-	-
SYS_C00205438	Check	"GAMEID" IS NOT NULL	-			-	-	ENABLED	09/23/2025 09:01:06 AM	-	-
SYS_C00205439	Check	"REFEREEID" IS NOT NULL	-			-	-	ENABLED	09/23/2025 09:01:06 AM	-	-
GAMEREFEREE_PK	Primary	-	-			GAMEREFEREEID	-	ENABLED	09/23/2025 09:01:06 AM	GAMEREFEREE_PK	-
GR_GAME_FK	Foreign	-	GR_GAME_FK (DBISYS5000GRP13_ISYS5000_2025_SEM2.GAME)			GAMEID	NO ACTION	ENABLED	09/23/2025 09:01:06 AM	-	-
GR_REFEREREE_FK	Foreign	-	GR_REFEREREE_FK (DBISYS5000GRP13_ISYS5000_2025_SEM2.REFEREE)			REFEREEID	NO ACTION	ENABLED	09/23/2025 02:26:02 PM	-	-

StartPosition
The SQL Syntax:

```
CREATE TABLE StartPosition (  
  
    positionID CHAR(4) NOT NULL,  
  
    description VARCHAR2(25) NOT NULL,  
  
    CONSTRAINT StartPosition_PK PRIMARY KEY (positionID)  
  
);
```

Result of syntax in Oracle:

STARTPOSITION

+

v

Table

Data

Indexes

Model

Constraints

Grants

Statistics

UI Defaults

Triggers

Dependencies

SQL

REST

Add Column

Modify Column

Rename Column

Drop Column

Rename

Copy

Drop

Truncate

Create Lookup Table

Column Name	Data Type	Nullable	Default	Primary Key
POSITIONID	CHAR(4)	No	-	1
DESCRIPTION	VARCHAR2(25)	No	-	-

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STARTPOSITION												+ v
Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL	REST	
Create	Drop	Enable	Disable									
Constraint	Type	Search Condition	Related Constraint	Columns	Delete Rule	Status	Last Change	Index	Invalid			
SYS_C00205328	Check	"POSITIONID" IS NOT NULL	-	-	-	ENABLED	09/23/2025 07:07:11 AM	-	-			
SYS_C00205329	Check	"DESCRIPTION" IS NOT NULL	-	-	-	ENABLED	09/23/2025 07:07:11 AM	-	-			
STARTPOSITION_PK	Primary	-	-	POSITIONID	-	ENABLED	09/23/2025 07:07:11 AM	STARTPOSITION_PK	-			

BestOnGround

The SQL Syntax:

```
CREATE TABLE BestOnGround (  
  
    bogID CHAR(3) NOT NULL,  
  
    rank CHAR(1) NOT NULL,  
  
    points NUMBER(3) NOT NULL,  
  
    CONSTRAINT BestOnGround_PK PRIMARY KEY (bestOnGroundID)  
  
);
```

Result of syntax in Oracle:

BESTONGROUND

+ ▼

Table

Data

Indexes

Model

Constraints

Grants

Statistics

UI Defaults

Triggers

Depen

Add Column

Modify Column

Rename Column

Drop Column

Rename

Copy

Drop

Truncate

Create Lookup Table

Column Name	Data Type	Nullable	Default	Primary Key
BOGID	CHAR(3)	No	-	1
RANK	CHAR(1)	No	-	-
POINTS	NUMBER(3,0)	No	-	-

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```

resultID CHAR(4) NOT NULL,

description VARCHAR2(15) NOT NULL,

points NUMBER(1,0) NOT NULL,

CONSTRAINT Result_PK PRIMARY KEY (resultID)

);

```

Result of syntax in Oracle:

RESULT

+ v

Table

Data

Indexes

Model

Constraints

Grants

Statistics

UI Defaults

Triggers

Dependencies

SQL

REST

Add Column

Modify Column

Rename Column

Drop Column

Rename

Copy

Drop

Truncate

Create Lookup Table

Column Name	Data Type	Nullable	Default	Primary Key
RESULTID	CHAR(4)	No	-	1
DESCRIPTION	VARCHAR2(15)	No	-	-
POINTS	NUMBER(1,0)	No	-	-

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RESULT										
Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL
Create Drop Enable Disable										
Constraint	Type	Search Condition	Related Constraint	Columns	Delete Rule	Status	Last Change	Index	Invalid	
SYS_C00205346	Check	"RESULTID" IS NOT NULL	-	-	-	ENABLED	09/23/2025 08:32:15 AM	-	-	
SYS_C00205347	Check	"DESCRIPTION" IS NOT NULL	-	-	-	ENABLED	09/23/2025 08:32:15 AM	-	-	
SYS_C00205348	Check	"POINTS" IS NOT NULL	-	-	-	ENABLED	09/23/2025 08:32:15 AM	-	-	
RESULT_PK	Primary	-	-	RESULTID	-	ENABLED	09/23/2025 08:32:15 AM	RESULT_PK	-	

Club

```

CREATE TABLE Club (

clubID CHAR(4) NOT NULL,

clubName VARCHAR2(25) NOT NULL,

yearEstablished NUMBER(4,0),

majorSponsor VARCHAR2(25),

numOfMembers NUMBER(5,0),

coachID CHAR(4) NOT NULL,

CONSTRAINT Club_PK PRIMARY KEY (clubID),

CONSTRAINT Club_Coach_FK FOREIGN KEY (coachID) REFERENCES Coach(coachID)

);

```

Result of syntax in Oracle:

CLUB

+ v

Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL	REST
Add Column	Modify Column	Rename Column	Drop Column	Rename	Copy	Drop	Truncate	Create Lookup Table			
Column Name			Data Type			Nullable		Default		Primary Key	
CLUBID			CHAR(4)			No		-		1	
CLUBNAME			VARCHAR2(25)			No		-		-	
YEARESTABLISHED			NUMBER(4,0)			Yes		-		-	
MAJORSPONSOR			VARCHAR2(25)			Yes		-		-	
NUMOFMEMBERS			NUMBER(5,0)			Yes		-		-	
COACHID			CHAR(4)			No		-		-	

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CLUB											+ v
Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL	REST
Create	Drop	Enable	Disable								
Constraint	Type	Search Condition	Related Constraint		Columns	Delete Rule	Status	Last Change	Index	Invalid	
SYS_C00205350	Check	"CLUBID" IS NOT NULL	-		-	-	ENABLED	09/23/2025 08:36:13 AM	-	-	
SYS_C00205351	Check	"CLUBNAME" IS NOT NULL	-		-	-	ENABLED	09/23/2025 08:36:13 AM	-	-	
SYS_C00205352	Check	"COACHID" IS NOT NULL	-		-	-	ENABLED	09/23/2025 08:36:13 AM	-	-	
CLUB_PK	Primary	-	-		CLUBID	-	ENABLED	09/23/2025 08:36:13 AM	CLUB_PK	-	
CLUB_COACH_FK	Foreign	-	CLUB_COACH_FK (DBISYS5000GRP13_ISYS5000_2025_SEM2.COACH)		COACHID	NO ACTION	ENABLED	09/23/2025 08:36:13 AM	-	-	

Player
The SQL Syntax:

```
CREATE TABLE Player (  
  
    playerID CHAR(5) NOT NULL,  
  
    firstName VARCHAR2(30) NOT NULL,  
  
    lastName VARCHAR2(30) NOT NULL,  
  
    phoneNum CHAR(10),  
  
    email VARCHAR2(30),  
  
    jerseyNumber VARCHAR2(2),  
  
    dateOfBirth DATE,  
  
    height NUMBER(3,0),  
  
    weight NUMBER(3,0),  
  
    clubID CHAR(4) NOT NULL,
```

CONSTRAINT Player_PK **PRIMARY KEY** (playerID),

CONSTRAINT Player_Club_FK **FOREIGN KEY** (clubID) **REFERENCES** Club(clubID)

);

Result of syntax in Oracle:

PLAYER

+ v

Table

Data

Indexes

Model

Constraints

Grants

Statistics

UI Defaults

Triggers

Dependencies

SQL

REST

Add Column

Modify Column

Rename Column

Drop Column

Rename

Copy

Drop

Truncate

Create Lookup Table

Column Name	Data Type	Nullable	Default	Primary Key
PLAYERID	CHAR(5)	No	-	1
FIRSTNAME	VARCHAR2(30)	No	-	-
LASTNAME	VARCHAR2(30)	No	-	-
PHONENUM	CHAR(10)	Yes	-	-
EMAIL	VARCHAR2(30)	Yes	-	-
JERSEYNUMBER	VARCHAR2(2)	Yes	-	-
DATEOFBIRTH	DATE	Yes	-	-
HEIGHT	NUMBER(3,0)	Yes	-	-
WEIGHT	NUMBER(3,0)	Yes	-	-
CLUBID	CHAR(4)	No	-	-

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PLAYER											+ v	
Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL	REST	
Create	Drop	Enable	Disable									
Constraint	Type	Search Condition	Related Constraint		Columns	Delete Rule	Status	Last Change	Index	Invalid		
PLAYER_CLUB_FK	Foreign	-	PLAYER_CLUB_FK (DBISYS5000GRP13_ISYS5000_2025_SEM2.CLUB)		CLUBID	NO ACTION	ENABLED	09/23/2025 08:39:14 AM	-	-		
SYS_C00205359	Check	"PLAYERID" IS NOT NULL	-		-	-	ENABLED	09/23/2025 08:39:14 AM	-	-		
SYS_C00205360	Check	"FIRSTNAME" IS NOT NULL	-		-	-	ENABLED	09/23/2025 08:39:14 AM	-	-		
SYS_C00205361	Check	"LASTNAME" IS NOT NULL	-		-	-	ENABLED	09/23/2025 08:39:14 AM	-	-		
SYS_C00205362	Check	"CLUBID" IS NOT NULL	-		-	-	ENABLED	09/23/2025 08:39:14 AM	-	-		
PLAYER_PK	Primary	-	-		PLAYERID	-	ENABLED	09/23/2025 08:39:14 AM	PLAYER_PK	-		

Ground

The SQL Syntax:

CREATE TABLE Ground (

groundID **CHAR**(4) **NOT NULL**,

name **VARCHAR2**(30) **NOT NULL**,

```
address VARCHAR2(40) NOT NULL,

capacity NUMBER(5,0) NOT NULL,

clubID CHAR(4) NOT NULL,

CONSTRAINT Ground_PK PRIMARY KEY (groundID),

CONSTRAINT Ground_Club_FK FOREIGN KEY (clubID) REFERENCES Club(clubID)

);
```

Result of syntax in Oracle:

GROUND											+ v			
Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL	REST			
Add Column Modify Column Rename Column Drop Column Rename Copy Drop Truncate Create Lookup Table														
Column Name				Data Type			Nullable		Default		Primary Key			
GROUNDID				CHAR(4)			No		-		1			
NAME				VARCHAR2(30)			No		-		-			
ADDRESS				VARCHAR2(25)			No		-		-			
CAPACITY				NUMBER(5,0)			No		-		-			
CLUBID				CHAR(4)			No		-		-			
Download Print														
GROUND											+ v			
Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL	REST			
Create Drop Enable Disable														
Constraint	Type	Search Condition	Related Constraint					Columns	Delete Rule	Status	Last Change	Index	Invalid	
SYS_C00205365	Check	"GROUNDID" IS NOT NULL	-						-	-	ENABLED	09/23/2025 08:39:49 AM	-	-
SYS_C00205366	Check	"NAME" IS NOT NULL	-						-	-	ENABLED	09/23/2025 08:39:49 AM	-	-
SYS_C00205367	Check	"ADDRESS" IS NOT NULL	-						-	-	ENABLED	09/23/2025 08:39:49 AM	-	-
SYS_C00205368	Check	"CAPACITY" IS NOT NULL	-						-	-	ENABLED	09/23/2025 08:39:49 AM	-	-
SYS_C00205369	Check	"CLUBID" IS NOT NULL	-						-	-	ENABLED	09/23/2025 08:39:49 AM	-	-
GROUND_PK	Primary	-	-						GROUNDID	-	ENABLED	09/23/2025 08:39:49 AM	GROUND_PK	-
GROUND_CLUB_FK	Foreign	-	-	GROUND_CLUB_FK (DBISYS5000GRP13_ISYS5000_2025_SEM2.CLUB)					CLUBID	NO ACTION	ENABLED	09/23/2025 08:39:49 AM	-	-

Game
The SQL Syntax:

```
CREATE TABLE Game (

    gameId CHAR(4) NOT NULL,
```

```
description VARCHAR2(15) NOT NULL,  
gameDate DATE NOT NULL,  
startTime DATE NOT NULL,  
endTime DATE NOT NULL,  
weekID CHAR(4) NOT NULL,  
groundID CHAR(4) NOT NULL,  
  
CONSTRAINT Game_PK PRIMARY KEY (gameID),  
  
CONSTRAINT Game_CompetitionWeek_FK FOREIGN KEY (weekID) REFERENCES  
CompetitionWeek(weekID),  
  
CONSTRAINT Game_Ground_FK FOREIGN KEY (groundID) REFERENCES Ground(groundID)  
);
```

Result of syntax in Oracle:

GAME										
Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL
Create Drop Enable Disable										
Constraint	Type	Search Condition	Related Constraint			Columns	Delete Rule	Status	Last Change	Index
GAME_COMPETITIONWEEK_FK	Foreign	-	GAME_COMPETITIONWEEK_FK (DBISYS5000GRP13_JSYS5000_2025_SEM2.COMPETITIONWEEK)			WEEKID	NO ACTION	ENABLED	09/23/2025 08:55:15 AM	-
SYS_C00205393	Check	"DESCRIPTION" IS NOT NULL	-			-	-	ENABLED	09/23/2025 08:55:15 AM	-
SYS_C00205394	Check	"GAMEDATE" IS NOT NULL	-			-	-	ENABLED	09/23/2025 08:55:15 AM	-
SYS_C00205395	Check	"STARTTIME" IS NOT NULL	-			-	-	ENABLED	09/23/2025 08:55:15 AM	-
SYS_C00205396	Check	"ENDTIME" IS NOT NULL	-			-	-	ENABLED	09/23/2025 08:55:15 AM	-
SYS_C00205397	Check	"WEEKID" IS NOT NULL	-			-	-	ENABLED	09/23/2025 08:55:15 AM	-
SYS_C00205398	Check	"GROUNDID" IS NOT NULL	-			-	-	ENABLED	09/23/2025 08:55:15 AM	-
GAME_PK	Primary	-	-			GAMEID	-	ENABLED	09/23/2025 08:55:15 AM	GAME_PK
GAME_GROUND_FK	Foreign	-	GAME_GROUND_FK (DBISYS5000GRP13_JSYS5000_2025_SEM2.GROUND)			GROUNDID	NO ACTION	ENABLED	09/23/2025 08:55:15 AM	-
SYS_C00205392	Check	"GAMEID" IS NOT NULL	-			-	-	ENABLED	09/23/2025 08:55:15 AM	-

ClubGame

The SQL Syntax:

```
CREATE TABLE ClubGame (
    clubGameID CHAR(5) NOT NULL,
    gameID CHAR(4) NOT NULL,
    clubID CHAR(4) NOT NULL,
    goalScored NUMBER(2,0) NOT NULL,
    resultID CHAR(4) NOT NULL,
    CONSTRAINT clubGame_PK PRIMARY KEY (clubGameID),
    CONSTRAINT clubGame_game_FK FOREIGN KEY (gameID) REFERENCES Game(gameID),
    CONSTRAINT clubGame_club_FK FOREIGN KEY (clubID) REFERENCES Club(clubID),
    CONSTRAINT clubGame_result_FK FOREIGN KEY (resultID) REFERENCES Result(resultID)
);
```

Result of syntax in Oracle:

CLUBGAME

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Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL	REST
Add Column	Modify Column	Rename Column	Drop Column	Rename	Copy	Drop	Truncate	Create Lookup Table			
Column Name	Data Type	Nullable	Default	Primary Key							
CLUBGAMEID	CHAR(4)	No	-	1							
GAMEID	CHAR(4)	No	-	-							
CLUBID	CHAR(4)	No	-	-							
GOALSCORED	NUMBER(2,0)	No	-	-							
RESULTID	CHAR(4)	No	-	-							

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CLUBGAME

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Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL	REST
Create	Drop	Enable	Disable								
Constraint	Type	Search Condition	Related Constraint	Columns	Delete Rule	Status	Last Change	Index	Invalid		
SYS_C00205443	Check	"CLUBGAMEID" IS NOT NULL	-	-	-	ENABLED	09/23/2025 09:05:37 AM	-	-		
SYS_C00205444	Check	"GAMEID" IS NOT NULL	-	-	-	ENABLED	09/23/2025 09:05:37 AM	-	-		
SYS_C00205445	Check	"CLUBID" IS NOT NULL	-	-	-	ENABLED	09/23/2025 09:05:37 AM	-	-		
SYS_C00205446	Check	"GOALSCORED" IS NOT NULL	-	-	-	ENABLED	09/23/2025 09:05:37 AM	-	-		
SYS_C00205447	Check	"RESULTID" IS NOT NULL	-	-	-	ENABLED	09/23/2025 09:05:37 AM	-	-		
CLUBGAME_PK	Primary	-	-	CLUBGAMEID	-	ENABLED	09/23/2025 09:05:37 AM	CLUBGAME_PK	-		
CLUBGAME_GAME_FK	Foreign	-	CLUBGAME_GAME_FK (DBISYS5000GRP13_ISYS5000_2025_SEM2.GAME)	GAMEID	NO ACTION	ENABLED	09/23/2025 09:05:37 AM	-	-		
CLUBGAME_CLUB_FK	Foreign	-	CLUBGAME_CLUB_FK (DBISYS5000GRP13_ISYS5000_2025_SEM2.CLUB)	CLUBID	NO ACTION	ENABLED	09/23/2025 09:05:37 AM	-	-		
CLUBGAME_RESULT_FK	Foreign	-	CLUBGAME_RESULT_FK (DBISYS5000GRP13_ISYS5000_2025_SEM2.RESULT)	RESULTID	NO ACTION	ENABLED	09/23/2025 09:05:37 AM	-	-		

GameReferee

The SQL Syntax:

```
CREATE TABLE GameReferee (  
  
    gameRefereeID CHAR(6) NOT NULL,  
  
    refRole VARCHAR2(15) NOT NULL,  
  
    gameID CHAR(4) NOT NULL,  
  
    refereeID CHAR(4) NOT NULL,  
  
    CONSTRAINT GameReferee_PK PRIMARY KEY (gameRefereeID),  
  
    CONSTRAINT GR_Game_FK FOREIGN KEY (gameID) REFERENCES Game(gameID),  
  
    CONSTRAINT GR_Referee_FK FOREIGN KEY (refereeID) REFERENCES Referee(refereeID)  
  
);
```

Result of syntax in Oracle:

GAMEREFEREE										+ ~	
Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL	REST
Add Column Modify Column Rename Column Drop Column Rename Copy Drop Truncate Create Lookup Table											
Column Name		Data Type		Nullable		Default		Primary Key			
GAMEREFEREEID		CHAR(6)		No		-		1			
REFROLE		VARCHAR2(15)		No		-		-			
GAMEID		CHAR(4)		No		-		-			
REFEREEID		CHAR(4)		No		-		-			
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GAMEREFEREE

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TableDataIndexesModelConstraints

GrantsStatisticsUI DefaultsTriggersDependenciesSQLREST

CreateDropEnableDisable

Constraint	Type	Search Condition	Related Constraint	Columns	Delete Rule	Status	Last Change	Index	Invalid
SYS_C00205436	Check	"GAMEREFEREEID" IS NOT NULL	-	-	-	ENABLED	09/23/2025 09:01:06 AM	-	-
SYS_C00205437	Check	"REFROLE" IS NOT NULL	-	-	-	ENABLED	09/23/2025 09:01:06 AM	-	-
SYS_C00205438	Check	"GAMEID" IS NOT NULL	-	-	-	ENABLED	09/23/2025 09:01:06 AM	-	-
SYS_C00205439	Check	"REFEREEID" IS NOT NULL	-	-	-	ENABLED	09/23/2025 09:01:06 AM	-	-
GAMEREFEREE_PK	Primary	-	-	GAMEREFEREEID	-	ENABLED	09/23/2025 09:01:06 AM	GAMEREFEREE_PK	-
GR_GAME_FK	Foreign	-	GR_GAME_FK (DBISYS5000GRP13_ISYS5000_2025_SEM2.GAME)	GAMEID	NO ACTION	ENABLED	09/23/2025 09:01:06 AM	-	-
GR_REFEREE_FK	Foreign	-	GR_REFEREE_FK (DBISYS5000GRP13_ISYS5000_2025_SEM2.REFEREE)	REFEREEID	NO ACTION	ENABLED	09/23/2025 09:01:06 AM	-	-

GamePlayer
The SQL Syntax:

```
CREATE TABLE GamePlayer (  
  
    gamePlayerID CHAR(6) NOT NULL,  
  
    gameID CHAR(4) NOT NULL,  
  
    playerID CHAR(5) NOT NULL,  
  
    positionID CHAR(4) NOT NULL,  
  
    gameMinute NUMBER(3,0) NOT NULL,  
  
    CONSTRAINT GamePlayer_PK PRIMARY KEY (gamePlayerID),  
  
    CONSTRAINT GamePlayer_Game_FK FOREIGN KEY (gameID) REFERENCES Game(gameID),  
  
    CONSTRAINT GamePlayer_Player_FK FOREIGN KEY (playerID) REFERENCES Player(playerID),  
  
    CONSTRAINT GamePlayer_Position_FK FOREIGN KEY (positionID) REFERENCES  
StartPosition(positionID)  
);
```

Result of syntax in Oracle:

GAMEPLAYER

+ ▾

Table

Data

Indexes

Model

Constraints

Grants

Statistics

UI Defaults

Triggers

Dependencies

SQL

REST

Add Column

Modify Column

Rename Column

Drop Column

Rename

Copy

Drop

Truncate

Create Lookup Table

Column Name	Data Type	Nullable	Default	Primary Key
GAMEPLAYERID	CHAR(6)	No	-	1
GAMEID	CHAR(4)	No	-	-
PLAYERID	CHAR(5)	No	-	-
POSITIONID	CHAR(4)	No	-	-
GAMEMINUTE	NUMBER(3,0)	No	-	-

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GAMEPLAYER												
Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL	REST	
Create Drop Enable Disable												
Constraint	Type	Search Condition	Related Constraint					Columns	Delete Rule	Status	Last Change	Index
GAMEPLAYER_GAME_FK	Foreign	-	GAMEPLAYER_GAME_FK (DBISYS5000GRP13_ISYS5000_2025_SEM2.GAME)					GAMEID	NO ACTION	ENABLED	09/23/2025 09:09:22 AM	-
SYS_C00205452	Check	"GAMEPLAYERID" IS NOT NULL	-					-	-	ENABLED	09/23/2025 09:09:22 AM	-
SYS_C00205453	Check	"GAMEID" IS NOT NULL	-					-	-	ENABLED	09/23/2025 09:09:22 AM	-
SYS_C00205454	Check	"PLAYERID" IS NOT NULL	-					-	-	ENABLED	09/23/2025 09:09:22 AM	-
SYS_C00205455	Check	"POSITIONID" IS NOT NULL	-					-	-	ENABLED	09/23/2025 09:09:22 AM	-
SYS_C00205456	Check	"GAMEMINUTE" IS NOT NULL	-					-	-	ENABLED	09/23/2025 09:09:22 AM	-
GAMEPLAYER_PK	Primary	-	-					GAMEPLAYERID	-	ENABLED	09/23/2025 09:09:22 AM	GAMEPLAYER_PK
GAMEPLAYER_PLAYER_FK	Foreign	-	GAMEPLAYER_PLAYER_FK (DBISYS5000GRP13_ISYS5000_2025_SEM2.PLAYER)					PLAYERID	NO ACTION	ENABLED	09/23/2025 09:09:22 AM	-
GAMEPLAYER_POSITION_FK	Foreign	-	GAMEPLAYER_POSITION_FK (DBISYS5000GRP13_ISYS5000_2025_SEM2.STARTPOSITION)					POSITIONID	NO ACTION	ENABLED	09/23/2025 09:09:22 AM	-

Substitution
The SQL Syntax:

```
CREATE TABLE Substitution (  
  
    substitutionID CHAR(6) NOT NULL,  
  
    gamePlayerOutID CHAR(6) NOT NULL,  
  
    gamePlayerInID CHAR(6) NOT NULL,  
  
    minute NUMBER(3,0) NOT NULL,  
  
    reason VARCHAR2(30),  
  
);
```



```

    CONSTRAINT AwardedCard_GamePlayer_FK FOREIGN KEY (gamePlayerID) REFERENCES
GamePlayer(gamePlayerID),

    CONSTRAINT AwardedCard_GameReferee_FK FOREIGN KEY (gameRefereeID) REFERENCES
GameReferee(gameRefereeID)

);

```

Result of syntax in Oracle:

AWARDEDCARD										
Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL
Add Column Modify Column Rename Column Drop Column Rename Copy Drop Truncate Create Lookup Table										
Column Name	Data Type		Nullable	Default		Primary Key				
AWARDEDCARDID	CHAR(5)		No	-		1				
CARDDESCRIPTION	VARCHAR2(6)		No	-		-				
MINUTE	NUMBER(3,0)		No	-		-				
GAMEPLAYERID	CHAR(6)		No	-		-				
GAMEREFEREED	CHAR(6)		No	-		-				

AWARDEDCARD										
Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL
Create Drop Enable Disable										
Constraint	Type	Search Condition		Related Constraint		Columns	Delete Rule	Status	Last Change	Index
SYS_C00205468	Check	"AWARDEDCARDID" IS NOT NULL		-		-	-	ENABLED	09/23/2025 09:11:51 AM	-
SYS_C00205469	Check	"CARDDESCRIPTION" IS NOT NULL		-		-	-	ENABLED	09/23/2025 09:11:51 AM	-
SYS_C00205470	Check	"MINUTE" IS NOT NULL		-		-	-	ENABLED	09/23/2025 09:11:51 AM	-
SYS_C00205471	Check	"GAMEPLAYERID" IS NOT NULL		-		-	-	ENABLED	09/23/2025 09:11:51 AM	-
SYS_C00205472	Check	"GAMEREFEREED" IS NOT NULL		-		-	-	ENABLED	09/23/2025 09:11:51 AM	-
AWARDEDCARD_PK	Primary	-		-		AWARDEDCARDID	-	ENABLED	09/23/2025 09:11:51 AM	AWARDEDCARD_PK
AWARDEDCARD_GAMEPLAYER_FK	Foreign	-		AWARDEDCARD_GAMEPLAYER_FK (DBISYS5000GRP13_ISYS5000_2025_SEM2.GAMEPLAYER)		GAMEPLAYERID	NO ACTION	ENABLED	09/23/2025 09:11:51 AM	-
AWARDEDCARD_GAMEREFREE_FK	Foreign	-		AWARDEDCARD_GAMEREFREE_FK (DBISYS5000GRP13_ISYS5000_2025_SEM2.GAMEREFREE)		GAMEREFEREED	NO ACTION	ENABLED	09/23/2025 09:11:51 AM	-

ScoredGoal

The SQL Syntax:

```

CREATE TABLE ScoredGoal (

    goalID CHAR(5) NOT NULL,

    gamePlayerID CHAR(6) NOT NULL,

    gameMinute NUMBER(3,0) NOT NULL,

    CONSTRAINT ScoredGoal_PK PRIMARY KEY (goalID),

    CONSTRAINT ScoredGoal_GamePlayer_FK FOREIGN KEY (gamePlayerID) REFERENCES
GamePlayer(gamePlayerID)

);

```

Result of syntax in Oracle:

SCOREDGOAL

+

▾

Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL	REST
Add Column	Modify Column	Rename Column	Drop Column	Rename	Copy	Drop	Truncate	Create Lookup Table			
Column Name		Data Type		Nullable		Default		Primary Key			
GOALID		CHAR(5)		No		-		1			
GAMEPLAYERID		CHAR(6)		No		-		-			
GAMEMINUTE		NUMBER(3,0)		No		-		-			

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SCOREDGOAL											+ ▾
Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL	REST
Create	Drop	Enable	Disable								
Constraint	Type	Search Condition	Related Constraint			Columns	Delete Rule	Status	Last Change	Index	Invalid
SYS_C00205476	Check	"GOALID" IS NOT NULL	-			-	-	ENABLED	09/23/2025 09:12:26 AM	-	-
SYS_C00205477	Check	"GAMEPLAYERID" IS NOT NULL	-			-	-	ENABLED	09/23/2025 09:12:26 AM	-	-
SYS_C00205478	Check	"GAMEMINUTE" IS NOT NULL	-			-	-	ENABLED	09/23/2025 09:12:26 AM	-	-
SCOREDGOAL_PK	Primary	-	-			GOALID	-	ENABLED	09/23/2025 09:12:26 AM	SCOREDGOAL_PK	-
SCOREDGOAL_GAMEPLAYER_FK	Foreign	-	SCOREDGOAL_GAMEPLAYER_FK (DBISYS5000GRP13_JSYS5000_2025_SEM2.GAMEPLAYER)			GAMEPLAYERID	NO ACTION	ENABLED	09/23/2025 09:12:26 AM	-	-

BogPlayer
The SQL Syntax:

```
CREATE TABLE BOGPlayer (  
  
    bogPlayerID CHAR(6) NOT NULL,  
  
    bogID CHAR(3) NOT NULL,  
  
    gamePlayerID CHAR(6) NOT NULL,  
  
    CONSTRAINT BogPlayer_PK PRIMARY KEY (bogPlayerID),  
  
    CONSTRAINT BogPlayer_BOG_FK FOREIGN KEY (bogID) REFERENCES BestOnGround(bogID),  
  
    CONSTRAINT BogPlayer_GamePlayer_FK FOREIGN KEY (gamePlayerID) REFERENCES  
    GamePlayer(gamePlayerID)  
  
);
```

Result of syntax in Oracle:

BOGPLAYER

+ v

Table

Data

Indexes

Model

Constraints

Grants

Statistics

UI Defaults

Triggers

Dependencies

SQL

REST

Add Column

Modify Column

Rename Column

Drop Column

Rename

Copy

Drop

Truncate

Create Lookup Table

Column Name	Data Type	Nullable	Default	Primary Key
BOGPLAYERID	CHAR(6)	No	-	1
BOGID	CHAR(3)	No	-	-
GAMEPLAYERID	CHAR(6)	No	-	-

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BOGPLAYER												+ v
Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL	REST	
Create	Drop	Enable	Disable									
Constraint		Type	Search Condition	Related Constraint			Columns	Delete Rule	Status	Last Change	Index	Invalid
SYS_C00205495		Check	"BOGPLAYERID" IS NOT NULL	-			-	-	ENABLED	09/23/2025 09:41:06 AM	-	-
SYS_C00205496		Check	"BOGID" IS NOT NULL	-			-	-	ENABLED	09/23/2025 09:41:06 AM	-	-
SYS_C00205497		Check	"GAMEPLAYERID" IS NOT NULL	-			-	-	ENABLED	09/23/2025 09:41:06 AM	-	-
BOGPLAYER_PK		Primary	-	-			BOGPLAYERID	-	ENABLED	09/23/2025 09:41:06 AM	BOGPLAYER_PK	-
BOGPLAYER_BOG_FK		Foreign	-	BOGPLAYER_BOG_FK (DBISYS5000GRP13_ISYS5000_2025_SEM2.BESTONGROUND)			BOGID	NO ACTION	ENABLED	09/23/2025 09:41:06 AM	-	-
BOGPLAYER_GAMEPLAYER_FK		Foreign	-	BOGPLAYER_GAMEPLAYER_FK (DBISYS5000GRP13_ISYS5000_2025_SEM2.GAMEPLAYER)			GAMEPLAYERID	NO ACTION	ENABLED	09/23/2025 09:41:06 AM	-	-

3.2 INSERT data SQL Syntax

Coach

The SQL Syntax:

INSERT ALL

```
INTO Coach (coachID, firstName, lastName, phoneNum, email, dateOfBirth, gender, startDate  
INSERT ALL
```

```
INTO Coach (coachID, firstName, lastName, phoneNum, email, dateOfBirth, gender, startDate)
```

```
VALUES ('CH01','Robert','Taylor','0415789833','robert.t@email.com',TO_DATE('04/01/1968','MM/DD/  
YYYY'),'M',TO_DATE('01/15/2001','MM/DD/YYYY'))
```

```
INTO Coach (coachID, firstName, lastName, phoneNum, email, dateOfBirth, gender, startDate)
```

```
VALUES ('CH02','Emily','Clarke','0413942657','emily.c@email.com',TO_DATE('09/14/1970','MM/DD/  
YYYY'),'F',TO_DATE('06/09/2003','MM/DD/YYYY'))
```

```
INTO Coach (coachID, firstName, lastName, phoneNum, email, dateOfBirth, gender, startDate)
```

```
VALUES ('CH03','Anthony','Li','0413390691','anthony.li@email.com',TO_DATE('12/21/1969','MM/DD/  
YYYY'),'M',TO_DATE('02/20/2006','MM/DD/YYYY'))
```

...

```
INTO Coach (coachID, firstName, lastName, phoneNum, email, dateOfBirth, gender, startDate)
```

```
VALUES ('CH07','Peter','Johnson','0412374254','peter.j@email.com',TO_DATE('01/05/1971','MM/DD/  
YYYY'),'M',TO_DATE('03/28/2005','MM/DD/YYYY'))
```

```
INTO Coach (coachID, firstName, lastName, phoneNum, email, dateOfBirth, gender, startDate)
```

```
VALUES ('CH08','Amanda','Wright','0411161045','amanda.w@email.com',TO_DATE('10/20/1974','MM/DD/  
YYYY'),'F',TO_DATE('07/30/2012','MM/DD/YYYY'))
```

```
SELECT * FROM dual;
```

Result of syntax in Oracle:

COACHID	FIRSTNAME	LASTNAME	PHONENUM	EMAIL	DATEOFBIRTH	GENDER	STARTDATE
CH01	Robert	Taylor	0415789833	robert.t@email.com	04/01/1968	M	01/15/2001
CH02	Emily	Clarke	0413942657	emily.c@email.com	09/14/1970	F	06/09/2003
CH03	Anthony	Li	0413390691	anthony.li@email.com	12/21/1969	M	02/20/2006
CH04	Sarah	Hughes	0416205432	sarah.h@email.com	07/11/1972	F	11/05/2008
CH05	Michael	Connor	0411396559	michael.oc@email.com	02/18/1967	M	09/01/2002
CH06	Lisa	Nguyen	0416751701	lisa.nguyen@email.com	06/29/1973	F	04/12/2010
CH07	Peter	Johnson	0412374254	peter.j@email.com	01/05/1971	M	03/28/2005
CH08	Amanda	Wright	0411161045	amanda.w@email.com	10/20/1974	F	07/30/2012

8 rows returned in 0.00 seconds

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CompetitionWeek

The SQL Syntax:

INSERT ALL

```
INTO CompetitionWeek (weekID, description, startDate, endDate)
```

```
VALUES ('WK01','Opening Round',TO_DATE('05/12/2025','MM/DD/YYYY'),TO_DATE('05/18/2025','MM/DD/YYYY'))
```

```

INSERT CompetitionWeek (weekID, description, startDate, endDate)

VALUES ('WK02','Rivalry Round',TO_DATE('05/19/2025','MM/DD/YYYY'),TO_DATE('05/25/2025','MM/DD/YYYY'))

INSERT CompetitionWeek (weekID, description, startDate, endDate)

VALUES ('WK03','Home Advantage Round',TO_DATE('05/26/2025','MM/DD/YYYY'),TO_DATE('06/01/2025','MM/DD/YYYY'))

...

VALUES ('WK12','Penultimate Round',TO_DATE('07/28/2025','MM/DD/YYYY'),TO_DATE('08/03/2025','MM/DD/YYYY'))

INSERT CompetitionWeek (weekID, description, startDate, endDate)

VALUES ('WK13','Climactic Round',TO_DATE('08/04/2025','MM/DD/YYYY'),TO_DATE('08/10/2025','MM/DD/YYYY'))

INSERT CompetitionWeek (weekID, description, startDate, endDate)

VALUES ('WK14','Final Matchweek',TO_DATE('08/11/2025','MM/DD/YYYY'),TO_DATE('08/17/2025','MM/DD/YYYY'))

SELECT * FROM dual;

```

Result of syntax in Oracle:

WEEKID	DESCRIPTION	STARTDATE	ENDDATE
WK01	Opening Round	05/12/2025	05/18/2025
WK02	Rivalry Round	05/19/2025	05/25/2025
WK03	Home Advantage Round	05/26/2025	06/01/2025
WK04	Mid-Season Clash	06/02/2025	06/08/2025
....			
WK11	Final Push Round	07/21/2025	07/27/2025
WK12	Penultimate Round	07/28/2025	08/03/2025
WK13	Climactic Round	08/04/2025	08/10/2025
WK14	Final Matchweek	08/11/2025	08/17/2025

14 rows returned in 0.01 seconds

[Download](#)

Referee

The SQL Syntax:

```
INSERT ALL
```

```
INSERT Referee (refereeID, firstName, lastName, dateOfBirth, phoneNum, startDate, gender, refLevel)
```

```
VALUES
```

```
('RF01','Sarah','Wilson',TO_DATE('06/12/1984','MM/DD/YYYY'),'0404000001',TO_DATE('03/01/2010','MM/DD/YYYY'),'F',1)
```

```
INSERT Referee (refereeID, firstName, lastName, dateOfBirth, phoneNum, startDate, gender, refLevel)
```

VALUES

```
('RF02','Michael','Jones',TO_DATE('11/22/1981','MM/DD/YYYY'),'0404000002',TO_DATE('05/10/2008','MM/DD/YYYY'),'M',1)
```

...

```
INTO Referee (refereeID, firstName, lastName, dateOfBirth, phoneNum, startDate, gender, refLevel)
```

VALUES

```
('RF11','Grace','Martin',TO_DATE('11/06/1994','MM/DD/YYYY'),'0404000011',TO_DATE('04/10/2019','MM/DD/YYYY'),'F',4)
```

```
INTO Referee (refereeID, firstName, lastName, dateOfBirth, phoneNum, startDate, gender, refLevel)
```

VALUES

```
('RF12','Thomas','Baker',TO_DATE('02/23/1979','MM/DD/YYYY'),'0404000012',TO_DATE('09/18/2006','MM/DD/YYYY'),'M',3)
```

```
SELECT * FROM dual;
```

Result of syntax in Oracle:

Results

Explain

Describe

Saved SQL

History

REFREEID	FIRSTNAME	LASTNAME	DATEOFBIRTH	PHONENUM	STARTDATE	GENDER	REFLEVEL
RF01	Sarah	Wilson	06/12/1984	0404000001	03/01/2010	F	1
RF02	Michael	Jones	11/22/1981	0404000002	05/10/2008	M	1
RF03	Linda	Tran	03/15/1988	0404000003	07/20/2015	F	2
RF04	James	Patel	09/30/1983	0404000004	02/14/2011	M	1
....							
RF10	William	Taylor	10/17/1988	0404000010	03/02/2014	M	2
RF11	Grace	Martin	11/06/1994	0404000011	04/10/2019	F	4
RF12	Thomas	Baker	02/23/1979	0404000012	09/18/2006	M	3

12 rows returned in 0.00 seconds

Download

StartPosition

The SQL Syntax:

```
INSERT ALL
```

```
INTO StartPosition (positionID, description) VALUES ('SP01','Goalkeeper')
```

```
INTO StartPosition (positionID, description) VALUES ('SP02','Right Back')
```

```
INTO StartPosition (positionID, description) VALUES ('SP03','Left Back')
```

```
INTO StartPosition (positionID, description) VALUES ('SP10','Left Wing')
```

```
INTO StartPosition (positionID, description) VALUES ('SP11','Center Forward')
```

```
INTO StartPosition (positionID, description) VALUES ('SP12','Substitute')
```

```
SELECT * FROM dual;
```

Result of syntax in Oracle:

POSITIONID	DESCRIPTION
SP01	Goalkeeper
SP02	Right Back
SP03	Left Back
SP04	Center Back
....	
SP09	Right Wing
SP10	Left Wing
SP11	Center Forward
SP12	Substitute

12 rows returned in 0.00 seconds [Download](#)

BestOnGround
The SQL Syntax:

```
INSERT ALL
INTO BestOnGround (bestOnGroundID, rank, points) VALUES ('BG1', 1, 3)
INTO BestOnGround (bestOnGroundID, rank, points) VALUES ('BG2', 2, 2)
INTO BestOnGround (bestOnGroundID, rank, points) VALUES ('BG3', 3, 1)
SELECT * FROM dual;
```

Result of syntax in Oracle:

BOGID	RANK	POINTS
BG1	1	3
BG2	2	2
BG3	3	1

3 rows returned in 0.00 seconds [Download](#)

Result
The SQL Syntax:

```
INSERT ALL
INTO Result (resultID, description, points) VALUES ('RS01','Away Win',6)
INTO Result (resultID, description, points) VALUES ('RS02','Home Win',3)
INTO Result (resultID, description, points) VALUES ('RS03','Away Draw',2)
INTO Result (resultID, description, points) VALUES ('RS04','Home Draw',1)
INTO Result (resultID, description, points) VALUES ('RS05','Away Loss',0)
INTO Result (resultID, description, points) VALUES ('RS06','Home Loss',0)
```

```
SELECT * FROM dual;
```

Result of syntax in Oracle:

RESULTID	DESCRIPTION	POINTS
RS01	Away Win	6
RS02	Home Win	3
RS03	Away Draw	2
RS04	Home Draw	1
RS05	Away Loss	0
RS06	Home Loss	0

6 rows returned in 0.01 seconds

[Download](#)

Club

The SQL Syntax:

```
INSERT ALL
```

```
INTO Club (clubID, clubName, yearEstablished, majorSponsor, numOfMembers, coachID)
```

```
VALUES ('CL01','Eagle Bay Eagles',1985,'Busselton Motors',120,'CH01')
```

```
INTO Club (clubID, clubName, yearEstablished, majorSponsor, numOfMembers, coachID)
```

```
VALUES ('CL02','Dunsborough Dingos',1978,'SurfCo Insurance',115,'CH02')
```

```
INTO Club (clubID, clubName, yearEstablished, majorSponsor, numOfMembers, coachID)
```

```
VALUES ('CL03','Broadwater Bandicoots',1990,'Busselton Fresh',130,'CH03')
```

```
INTO Club (clubID, clubName, yearEstablished, majorSponsor, numOfMembers, coachID)
```

```
VALUES ('CL04','Mary-Brook Magpies',1982,'Brook Wines',110,'CH04')
```

```
INTO Club (clubID, clubName, yearEstablished, majorSponsor, numOfMembers, coachID)
```

```
VALUES ('CL05','Reinscourt Rabbits',1988,'West Coast Bank',125,'CH05')
```

```
INTO Club (clubID, clubName, yearEstablished, majorSponsor, numOfMembers, coachID)
```

```
VALUES ('CL06','Wonnerup Wombats',1975,'Wonn Chemicals',118,'CH06')
```

```
INTO Club (clubID, clubName, yearEstablished, majorSponsor, numOfMembers, coachID)
```

```
VALUES ('CL07','New Caves Rd Numbats',1993,'Cave Tours WA',112,'CH07')
```

```
INTO Club (clubID, clubName, yearEstablished, majorSponsor, numOfMembers, coachID)
```

```
VALUES ('CL08','Hanes Hawkes',1980,'Hanes Builders',128,'CH08')
```

```
SELECT * FROM DUAL;
```

Result of syntax in Oracle:

CLUBID	CLUBNAME	YEARESTABLISHED	MAJORSPONSOR	NUMOFMEMBERS	COACHID
CL01	Eagle Bay Eagles	1985	Busselton Motors	120	CH01
CL02	Dunsborough Dingos	1978	SurfCo Insurance	115	CH02
CL03	Broadwater Bandicoots	1990	Busselton Fresh	130	CH03
CL04	Mary-Brook Magpies	1982	Brook Wines	110	CH04
CL05	Reinscourt Rabbits	1988	West Coast Bank	125	CH05
CL06	Wonnerup Wombats	1975	Wonn Chemicals	118	CH06
CL07	New Caves Rd Numbats	1993	Cave Tours WA	112	CH07
CL08	Hanes Hawkes	1980	Hanes Builders	128	CH08

8 rows returned in 0.00 seconds [Download](#)

Player

The SQL Syntax:

INSERT ALL

INTO Player (playerID, firstName, lastName, phoneNum, email, jerseyNumber, dateOfBirth, height, **weight**, clubID)

VALUES ('PL101','Liam','Taylor','0495888726','liam.taylor1@email.com','80',TO_DATE('10-06-98','DD-MM-RR'),175,75,'CL01')

INTO Player (playerID, firstName, lastName, phoneNum, email, jerseyNumber, dateOfBirth, height, **weight**, clubID)

VALUES ('PL102','Martin','Carter','0422519875','martin.carter2@email.com','60',TO_DATE('04-03-00','DD-MM-RR'),187,79,'CL01')

INTO Player (playerID, firstName, lastName, phoneNum, email, jerseyNumber, dateOfBirth, height, **weight**, clubID)

VALUES ('PL103','Stuart','Martin','0416138399','stuart.martin3@email.com','98',TO_DATE('22-12-01','DD-MM-RR'),174,76,'CL01')

INTO Player (playerID, firstName, lastName, phoneNum, email, jerseyNumber, dateOfBirth, height, **weight**, clubID)

VALUES ('PL104','Mark','Wright','0427054234','mark.wright4@email.com','83',TO_DATE('20-04-92','DD-MM-RR'),177,79,'CL01')

INTO Player (playerID, firstName, lastName, phoneNum, email, jerseyNumber, dateOfBirth, height, **weight**, clubID)

....

VALUES ('PL809','George','Evans','0458920186','george.evans93@email.com','34',TO_DATE('20-10-96','DD-MM-RR'),187,82,'CL08')

INTO Player (playerID, firstName, lastName, phoneNum, email, jerseyNumber, dateOfBirth, height, **weight**, clubID)

VALUES ('PL810','Chris','Brooks','0425974901','chris.brooks94@email.com','3',TO_DATE('01-07-99','DD-MM-RR'),177,69,'CL08')

INTO Player (playerID, firstName, lastName, phoneNum, email, jerseyNumber, dateOfBirth, height, **weight**, clubID)

```
VALUES ('PL811','Daniel','Carter','0416421248','daniel.carter95@email.com','72',TO_DATE('26-08-97','DD-MM-RR'),190,80,'CL08')

INSERT INTO Player (playerID, firstName, lastName, phoneNum, email, jerseyNumber, dateOfBirth, height, weight, clubID)

VALUES ('PL812','Peter','Foster','0420056507','peter.foster96@email.com','16',TO_DATE('24-10-94','DD-MM-RR'),180,79,'CL08')

SELECT * FROM DUAL;
```

Result of syntax in Oracle:

Results

Explain

Describe

Saved SQL

History

PLAYERID	FIRSTNAME	LASTNAME	PHONENUM	EMAIL	JERSEYNUMBER	DATEOFBIRTH	HEIGHT	WEIGHT	CLUBID
PL101	Liam	Taylor	0495888726	liam.taylor1@email.com	80	06/10/1998	175	75	CL01
PL102	Martin	Carter	0422519875	martin.carter2@email.com	60	03/04/2000	187	79	CL01
PL103	Stuart	Martin	0416138399	stuart.martin3@email.com	98	12/22/2001	174	76	CL01
PL104	Mark	Wright	0427054234	mark.wright4@email.com	83	04/20/1992	177	79	CL01
PL105	Alan	Hughes	0465164748	alan.hughes5@email.com	49	05/15/1999	173	62	CL01
PL106	Martin	Wilson	0414826896	martin.wilson6@email.com	33	01/17/2004	184	80	CL01
...									
PL809	George	Evans	0458920186	george.evans93@email.com	34	10/20/1996	187	82	CL08
PL810	Chris	Brooks	0425974901	chris.brooks94@email.com	3	07/01/1999	177	69	CL08
PL811	Daniel	Carter	0416421248	daniel.carter95@email.com	72	08/26/1997	190	80	CL08
PL812	Peter	Foster	0420056507	peter.foster96@email.com	16	10/24/1994	180	79	CL08

96 rows returned in 0.01 seconds

[Download](#)

Ground The SQL Syntax:

```
INSERT ALL

INSERT INTO Ground (groundID, name, address, capacity, clubID)

VALUES ('GR01','Eagle Bay Centre','Eagle Bay Rd, Busselton WA',2000,'CL01')

INSERT INTO Ground (groundID, name, address, capacity, clubID)

VALUES ('GR02','Quedjinup Reserve','Naturaliste Terrace, WA',1500,'CL02')

INSERT INTO Ground (groundID, name, address, capacity, clubID)

VALUES ('GR03','Broadwater Bay Oval','Broadwater WA',2200,'CL03')

INSERT INTO Ground (groundID, name, address, capacity, clubID)

VALUES ('GR06','Wonnerup Sports Ground','Wonnerup WA',1700,'CL06')

INSERT INTO Ground (groundID, name, address, capacity, clubID)

VALUES ('GR07','Caves Road Reserve','Caves Rd, Busselton WA',1600,'CL07')

INSERT INTO Ground (groundID, name, address, capacity, clubID)
```

```
VALUES ('GR08','Hanes Sports Ground','Hanes WA',1900,'CL08')

SELECT * FROM dual;
```

Result of syntax in Oracle:

GROUNDID	NAME	ADDRESS	CAPACITY	CLUBID
GR01	Eagle Bay Centre	Eagle Bay Rd, Busselton WA	2000	CL01
GR02	Quedjinup Reserve	Naturaliste Terrace, WA	1500	CL02
GR03	Broadwater Bay Oval	Broadwater WA	2200	CL03
GR04	Kealy Reserve	Kealy WA	1800	CL04
GR05	West Busselton Oval	West Busselton WA	2100	CL05
GR06	Wonnerup Sports Ground	Wonnerup WA	1700	CL06
GR07	Caves Road Reserve	Caves Rd, Busselton WA	1600	CL07
GR08	Hanes Sports Ground	Hanes WA	1900	CL08

8 rows returned in 0.00 seconds [Download](#)

Game
The SQL Syntax:

```
INSERT ALL

INTO Game (GAMEID, DESCRIPTION, GAMEDATE, STARTTIME, ENDTIME, WEEKID, GROUNDID) VALUES
('GM01','CL01 v
CL02',TO_DATE('05/12/2025','MM/DD/YYYY'),TO_DATE('08:00','HH24:MI'),TO_DATE('09:30','HH24:MI'),'WK01','G
R01')

INTO Game (GAMEID, DESCRIPTION, GAMEDATE, STARTTIME, ENDTIME, WEEKID, GROUNDID) VALUES
('GM02','CL03 v
CL04',TO_DATE('05/14/2025','MM/DD/YYYY'),TO_DATE('14:00','HH24:MI'),TO_DATE('15:30','HH24:MI'),'WK01','G
R03')

INTO Game (GAMEID, DESCRIPTION, GAMEDATE, STARTTIME, ENDTIME, WEEKID, GROUNDID) VALUES
('GM03','CL05 v
CL06',TO_DATE('05/16/2025','MM/DD/YYYY'),TO_DATE('08:00','HH24:MI'),TO_DATE('09:30','HH24:MI'),'WK01','G
R05')

....

INTO Game (GAMEID, DESCRIPTION, GAMEDATE, STARTTIME, ENDTIME, WEEKID, GROUNDID) VALUES
('GM54','CL03 v
CL02',TO_DATE('08/13/2025','MM/DD/YYYY'),TO_DATE('14:00','HH24:MI'),TO_DATE('15:30','HH24:MI'),'WK14','G
R03')

INTO Game (GAMEID, DESCRIPTION, GAMEDATE, STARTTIME, ENDTIME, WEEKID, GROUNDID) VALUES
('GM55','CL07 v
CL04',TO_DATE('08/15/2025','MM/DD/YYYY'),TO_DATE('08:00','HH24:MI'),TO_DATE('09:30','HH24:MI'),'WK14','G
R07')

INTO Game (GAMEID, DESCRIPTION, GAMEDATE, STARTTIME, ENDTIME, WEEKID, GROUNDID) VALUES
('GM56','CL06 v
CL05',TO_DATE('08/17/2025','MM/DD/YYYY'),TO_DATE('14:00','HH24:MI'),TO_DATE('15:30','HH24:MI'),'WK14','G
R06')

SELECT * FROM dual;
```

**** The game table was later updated to insert data for numOfSpectators ****

```
BEGIN
UPDATE Game SET numOfSpectators = 1850 WHERE gameId = 'GM01';

UPDATE Game SET numOfSpectators = 1200 WHERE gameId = 'GM02';

UPDATE Game SET numOfSpectators = 1650 WHERE gameId = 'GM03';

UPDATE Game SET numOfSpectators = 950 WHERE gameId = 'GM04';

UPDATE Game SET numOfSpectators = 1920 WHERE gameId = 'GM05';

...

UPDATE Game SET numOfSpectators = 1280 WHERE gameId = 'GM52';

UPDATE Game SET numOfSpectators = 1640 WHERE gameId = 'GM53';

UPDATE Game SET numOfSpectators = 1980 WHERE gameId = 'GM54';

UPDATE Game SET numOfSpectators = 1020 WHERE gameId = 'GM55';

UPDATE Game SET numOfSpectators = 1420 WHERE gameId = 'GM56';

COMMIT;

EXCEPTION

WHEN OTHERS THEN

ROLLBACK;

RAISE;

END;
```

Result of syntax in Oracle:

Results

Explain

Describe

Saved SQL

History

GAMEID	DESCRIPTION	GAMEDATE	STARTTIME	ENDTIME	WEEKID	GROUNDID
GM01	CL01 v CL02	05/12/2025	09/01/2025	09/01/2025	WK01	GR01
GM02	CL03 v CL04	05/14/2025	09/01/2025	09/01/2025	WK01	GR03
GM03	CL05 v CL06	05/16/2025	09/01/2025	09/01/2025	WK01	GR05
GM04	CL07 v CL08	05/18/2025	09/01/2025	09/01/2025	WK01	GR07
GM05	CL01 v CL03	05/19/2025	09/01/2025	09/01/2025	WK02	GR01
...						
GM54	CL03 v CL02	08/13/2025	09/01/2025	09/01/2025	WK14	GR03
GM55	CL07 v CL04	08/15/2025	09/01/2025	09/01/2025	WK14	GR07
GM56	CL06 v CL05	08/17/2025	09/01/2025	09/01/2025	WK14	GR06

56 rows returned in 0.00 seconds

Download

ClubGame The SQL Syntax:

```
INSERT ALL
```

■ ■ ■

GRF166	Main	GM56	RF04
GRF167	Assistant1	GM56	RF05
GRF168	Assistant2	GM56	RF08

168 rows returned in 0.01 seconds

[Download](#)

GamePlayer

The SQL Syntax:

INSERT ALL

INTO GamePlayer (gamePlayerID, gameId, playerId, positionID, gameMinute)

VALUES ('GP0001','GM01','PL101','SP01',90)

INTO GamePlayer (gamePlayerID, gameId, playerId, positionID, gameMinute)

VALUES ('GP0002','GM01','PL102','SP02',90)

INTO GamePlayer (gamePlayerID, gameId, playerId, positionID, gameMinute)

VALUES ('GP0003','GM01','PL103','SP03',90)

INTO GamePlayer (gamePlayerID, gameId, playerId, positionID, gameMinute)

VALUES ('GP0004','GM01','PL104','SP04',90)

INTO GamePlayer (gamePlayerID, gameId, playerId, positionID, gameMinute)

VALUES ('GP0005','GM01','PL105','SP05',90)

...

INTO GamePlayer (gamePlayerID, gameId, playerId, positionID, gameMinute)

VALUES ('GP1340','GM56','PL408','SP08',90)

INTO GamePlayer (gamePlayerID, gameId, playerId, positionID, gameMinute)

VALUES ('GP1341','GM56','PL409','SP09',90)

INTO GamePlayer (gamePlayerID, gameId, playerId, positionID, gameMinute)

VALUES ('GP1342','GM56','PL410','SP10',90)

INTO GamePlayer (gamePlayerID, gameId, playerId, positionID, gameMinute)

VALUES ('GP1343','GM56','PL411','SP11',90)

INTO GamePlayer (gamePlayerID, gameId, playerId, positionID, gameMinute)

VALUES ('GP1344','GM56','PL412','SP12',9)

SELECT * FROM dual;

Result of syntax in Oracle:

Results	Explain	Describe	Saved SQL	History
GAMEPLAYERID	GAMEID	PLAYERID	POSITIONID	GAMEMINUTE
GP0001	GM01	PL101	SP01	90
GP0002	GM01	PL102	SP02	90
GP0003	GM01	PL103	SP03	90
GP0004	GM01	PL104	SP04	90
GP0005	GM01	PL105	SP05	90
...				
GP1342	GM56	PL410	SP10	90
GP1343	GM56	PL411	SP11	90
GP1344	GM56	PL412	SP12	0

1344 rows returned in 0.05 seconds [Download](#)

Substitution

The SQL Syntax:

INSERT ALL

INTO Substitution (substitutionID, gamePlayerOutID, gamePlayerInID, minute, reason) VALUES ('SUB001', 'GP0022', 'GP0024', 69, 'Fatigue')

INTO Substitution (substitutionID, gamePlayerOutID, gamePlayerInID, minute, reason) VALUES ('SUB002', 'GP0045', 'GP0048', 26, 'Performance issue')

INTO Substitution (substitutionID, gamePlayerOutID, gamePlayerInID, minute, reason) VALUES ('SUB003', 'GP0053', 'GP0060', 24, 'Injury concern')

INTO Substitution (substitutionID, gamePlayerOutID, gamePlayerInID, minute, reason) VALUES ('SUB004', 'GP0087', 'GP0096', 80, 'Fatigue')

....

INTO Substitution (substitutionID, gamePlayerOutID, gamePlayerInID, minute, reason) VALUES ('SUB053', 'GP1254', 'GP1260', 70, 'Injury concern')

INTO Substitution (substitutionID, gamePlayerOutID, gamePlayerInID, minute, reason) VALUES ('SUB054', 'GP1279', 'GP1284', 24, 'Tactical change')

INTO Substitution (substitutionID, gamePlayerOutID, gamePlayerInID, minute, reason) VALUES ('SUB055', 'GP1299', 'GP1308', 59, 'Strategic substitution')

INTO Substitution (substitutionID, gamePlayerOutID, gamePlayerInID, minute, reason) VALUES ('SUB056', 'GP1330', 'GP1332', 9, 'Fatigue')

SELECT * FROM dual;

Result of syntax in Oracle:

Results	Explain	Describe	Saved SQL	History
SUBSTITUTIONID	GAMEPLAYEROUTID	GAMEPLAYERINID	MINUTE	REASON
SUB001	GP0022	GP0024	69	Fatigue
SUB002	GP0045	GP0048	26	Performance issue
SUB003	GP0053	GP0060	24	Injury concern
SUB004	GP0087	GP0096	80	Fatigue
...				
SUB054	GP1279	GP1284	24	Tactical change
SUB055	GP1299	GP1308	59	Strategic substitution
SUB056	GP1330	GP1332	9	Fatigue

56 rows returned in 0.00 seconds [Download](#)

AwardedCard

The SQL Syntax:

INSERT ALL

INTO AwardedCard (awardedCardID, cardDescription, minute, gamePlayerID, gameRefereeID) VALUES ('AC001', 'Yellow', 15, 'GP0010', 'GRF001')

INTO AwardedCard (awardedCardID, cardDescription, minute, gamePlayerID, gameRefereeID) VALUES ('AC002', 'Yellow', 22, 'GP0009', 'GRF001')

INTO AwardedCard (awardedCardID, cardDescription, minute, gamePlayerID, gameRefereeID) VALUES ('AC003', 'Yellow', 25, 'GP0009', 'GRF001')

INTO AwardedCard (awardedCardID, cardDescription, minute, gamePlayerID, gameRefereeID) VALUES ('AC004', 'Yellow', 28, 'GP0025', 'GRF004')

....

INTO AwardedCard (awardedCardID, cardDescription, minute, gamePlayerID, gameRefereeID) VALUES ('AC012', 'Yellow', 76, 'GP0990', 'GRF124')

INTO AwardedCard (awardedCardID, cardDescription, minute, gamePlayerID, gameRefereeID) VALUES ('AC013', 'Yellow', 42, 'GP0995', 'GRF124')

INTO AwardedCard (awardedCardID, cardDescription, minute, gamePlayerID, gameRefereeID) VALUES ('AC014', 'Yellow', 74, 'GP0995', 'GRF124')

INTO AwardedCard (awardedCardID, cardDescription, minute, gamePlayerID, gameRefereeID) VALUES ('AC015', 'Red', 50, 'GP0220', 'GRF028')

SELECT * FROM dual;

Result of syntax in Oracle:

Results	Explain	Describe	Saved SQL	History
AWARDEDCARDID	CARDDESCRIPTION	MINUTE	GAMEPLAYERID	GAMEREFEREED
AC001	Yellow	15	GP0010	GRF001
AC002	Yellow	22	GP0009	GRF001
AC003	Yellow	25	GP0009	GRF001
AC004	Yellow	28	GP0025	GRF004
...				
AC012	Yellow	76	GP0990	GRF124
AC013	Yellow	42	GP0995	GRF124
AC014	Yellow	74	GP0995	GRF124
ACO15	Red	50	GP0220	GRF028

13 rows returned in 0.00 seconds [Download](#)

ScoredGoal

The SQL Syntax:

INSERT ALL

INTO ScoredGoal (goalID, gamePlayerID, gameMinute) VALUES ('GL001', 'GP0010', 44)

INTO ScoredGoal (goalID, gamePlayerID, gameMinute) VALUES ('GL002', 'GP0010', 66)

INTO ScoredGoal (goalID, gamePlayerID, gameMinute) VALUES ('GL003', 'GP0010', 78)

INTO ScoredGoal (goalID, gamePlayerID, gameMinute) VALUES ('GL004', 'GP0022', 40)

....

INTO ScoredGoal (goalID, gamePlayerID, gameMinute) VALUES ('GL150', 'GP1283', 89)

INTO ScoredGoal (goalID, gamePlayerID, gameMinute) VALUES ('GL151', 'GP1319', 65)

INTO ScoredGoal (goalID, gamePlayerID, gameMinute) VALUES ('GL152', 'GP1342', 77)

INTO ScoredGoal (goalID, gamePlayerID, gameMinute) VALUES ('GL153', 'GP1328', 83)

SELECT * FROM dual;

Result of syntax in Oracle:

Results	Explain	Describe	Saved SQL	History
GOALID	GAMEPLAYERID	GAMEMINUTE		
GL001	GP0010	44		
GL002	GP0010	66		
GL003	GP0010	78		
GL004	GP0022	40		
GL005	GP0011	89		

...

GL150	GP1283	89
GL151	GP1319	65
GL152	GP1342	77
GL153	GP1328	83

153 rows returned in 0.00 seconds [Download](#)

BogPlayer

The SQL Syntax:

INSERT ALL

INTO BogPlayer (bogPlayerID, bogID, gamePlayerID) VALUES ('BP0001', 'BG1', 'GP0010')

INTO BogPlayer (bogPlayerID, bogID, gamePlayerID) VALUES ('BP0002', 'BG2', 'GP0022')

INTO BogPlayer (bogPlayerID, bogID, gamePlayerID) VALUES ('BP0003', 'BG3', 'GP0011')

....

INTO BogPlayer (bogPlayerID, bogID, gamePlayerID) VALUES ('BP0166', 'BG1', 'GP1342')

INTO BogPlayer (bogPlayerID, bogID, gamePlayerID) VALUES ('BP0167', 'BG2', 'GP1328')

INTO BogPlayer (bogPlayerID, bogID, gamePlayerID) VALUES ('BP0168', 'BG3', 'GP1331')

SELECT * FROM dual;

Result of syntax in Oracle:

BOGPLAYERID	BOGID	GAMEPLAYERID
BP0001	BG1	GP0010
BP0002	BG2	GP0022
BP0003	BG3	GP0011
BP0004	BG1	GP0046

...

BP0165	BG3	GP1316
BP0166	BG1	GP1342
BP0167	BG2	GP1328
BP0168	BG3	GP1331

168 rows returned in 0.00 seconds [Download](#)

4. SQL Statements/Exercises

Write SQL scripts to test the functionality and effectiveness of the database you have created. You can include comments with your scripts, informing the purpose of the script, why you wrote it the way you did, who contributed and issues you encountered or problems you had to solve. These notes will be especially important if you can't get a script to work properly, as it will provide the only evidence of your thinking to the marker of your work.

Each Script should be presented in text – showing the SQL* used, underneath the appropriate heading (so it is obvious which question you are answering) followed by a screen-grab – using a

similar approach to the example demonstrated in figure 3 – followed by a description/explanation of the SQL used.

* NOTE: The SQL coding will be copied and tested by the marker. The coding must firstly actually work, and secondly answer the question properly, to gain marks.

4.1 Season Outcomes

4.1a Use the data in the various tables to find which two clubs would take part in a post-season game between the clubs that came first and second after 14 weeks of competition. You should include aggregated information such as total points scored, away and home results, total goals scored and the like.

Syntax for query 4.1a

```
SELECT

cl.clubName AS "Club Name",

crs.total_result_points AS "Result Points",

bogp.total_bog_points AS "BOG Points",

tgs.goals_scored_count AS "Goals Scored",

hws.home_win_count AS "Home Wins",

aws.away_win_count AS "Away Wins",

crs.total_result_points + bogp.total_bog_points AS "Total Points (Result + BOG)"

FROM Club cl

JOIN (

SELECT cg.clubID, SUM(rs.points) AS total_result_points

FROM ClubGame cg

JOIN Result rs ON rs.resultID = cg.resultID

GROUP BY cg.clubID

) crs ON crs.clubID = cl.clubID

JOIN (

SELECT pl.clubID, SUM(bog.points) AS total_bog_points

FROM BogPlayer bp

JOIN BestOnGround bog ON bog.bogID = bp.bogID

JOIN GamePlayer gp ON gp.gamePlayerID = bp.gamePlayerID

JOIN Player pl ON pl.playerID = gp.playerID

GROUP BY pl.clubID

) bogp ON bogp.clubID = cl.clubID

JOIN (

SELECT cg.clubID, SUM(cg.goalScored) AS goals_scored_count
```

```

FROM ClubGame cg

GROUP BY cg.clubID

) tgs ON tgs.clubID = cl.clubID

JOIN (

SELECT cg.clubID, COUNT(*) AS home_win_count

FROM ClubGame cg

WHERE cg.resultID = 'RS02'

GROUP BY cg.clubID

) hws ON hws.clubID = cl.clubID

JOIN (

SELECT cg.clubID, COUNT(*) AS away_win_count

FROM ClubGame cg

WHERE cg.resultID = 'RS01'

GROUP BY cg.clubID

) aws ON aws.clubID = cl.clubID

ORDER BY "Result Points" DESC;

```

Screen-grab/Result of query 4.1a

Results

Explain

Describe

Saved SQL

History

Club Name	Result Points	BOG Points	Goals Scored	Home Wins	Away Wins	Total Points (Result + BOG)
Wonnerup Wombats	38	45	20	3	4	83
Eagle Bay Eagles	37	46	22	5	3	83
Hanes Hawkes	36	43	17	2	4	79
Dunsborough Dingos	32	44	20	2	4	76
Broadwater Bandicoots	29	41	17	1	3	70
Mary-Brook Magpies	26	40	19	1	3	66
New Caves Rd Numbats	23	39	18	3	2	62
Reinscourt Rabbits	22	38	20	1	2	60

8 rows returned in 0.02 seconds

Download

8 rows returned in 0.02 seconds

[Download](#)

Explanation of query 4.1a

- The SELECT query has one main table which is Club (cl).
- The SQL query retrieves the clubName from the Club (cl) table.
- Subqueries are used to calculate additional information for each club:
 - From ClubGame (cg) and Result (rs) tables to SUM the result points as Result Points (alias crs).
 - From BogPlayer (bp), BestOnGround (bog), GamePlayer (gp) and Player (pl) tables to SUM the Best-on-Ground points as BOG Points (alias bogp).
 - From ClubGame (cg) table to SUM the goals scored as Goals Scored (alias tgs).

From ClubGame (cg) table, filtered where resultID = 'RS02', to COUNT the number of home wins as Home Wins (alias hws).

From ClubGame (cg) table, filtered where resultID = 'RS01', to COUNT the number of away wins as Away Wins (alias aws).

The query also calculates a Total Points (Result + BOG) column by adding together the result points and the BOG points, giving an overall score that can be used to compare clubs.

- The query uses aliases like cl, cg, rs, bp, bog, gp, and pl to make the SQL shorter and easier to read, instead of repeating full table names every time.
- Each subquery applies GROUP BY clubID so that all the rows for one club are grouped together, giving a single total for things like points or goals.
- The results from these subqueries are then joined back to the Club (cl) table using clubID, so that each club's name appears alongside its calculated totals.
- The final step uses ORDER BY "Result Points" DESC to sort the output, showing the clubs with the highest points at the top and making it clear which clubs are leading.

4.1b Produce a Competition ladder for the end of the home-and-away season of games that lists the clubs in order of where they finished (from 1st to 8th), how many competition points they earned, how many home games and away games they won and/or lost, and how many total number of goals they scored for the whole season

Syntax for query 4.1b

```
SELECT cl.clubID, cl.clubName,

(SELECT SUM(rs.points)

FROM ClubGame cg

JOIN Result rs ON cg.resultID = rs.resultID

WHERE cg.clubID = cl.clubID) AS "Total Competition Point",

(SELECT SUM(cg.goalScored)

FROM ClubGame cg

WHERE cg.clubID = cl.clubID) AS "Total Goal Score",

(SELECT COUNT(*)

FROM ClubGame cg

JOIN Result rs ON cg.resultID = rs.resultID

JOIN Game gm ON cg.gameID = gm.gameID

JOIN Ground grn ON gm.groundID = grn.groundID

WHERE cg.clubID = cl.clubID AND rs.description = 'Home Win') AS "Home Win",

(SELECT COUNT(*)

FROM ClubGame cg

JOIN Result rs ON cg.resultID = rs.resultID
```

```

JOIN Game gm ON cg.gameID = gm.gameID

JOIN Ground grn ON gm.groundID = grn.groundID

WHERE cg.clubID = cl.clubID AND rs.description = 'Home Loss') AS "Home Loss",

(SELECT COUNT(*)

FROM ClubGame cg

JOIN Result rs ON cg.resultID = rs.resultID

JOIN Game gm ON cg.gameID = gm.gameID

JOIN Ground grn ON gm.groundID = grn.groundID

WHERE cg.clubID = cl.clubID AND rs.description = 'Away Win') AS "Away Win",

(SELECT COUNT(*)

FROM ClubGame cg

JOIN Result rs ON cg.resultID = rs.resultID

JOIN Game gm ON cg.gameID = gm.gameID

JOIN Ground grn ON gm.groundID = grn.groundID

WHERE cg.clubID = cl.clubID AND rs.description = 'Away Loss') AS "Away Loss"

FROM Club cl

ORDER BY "Total Competition Point" DESC, "Total Goal Score" DESC;

```

Screen-grab/Result of query 4.1b

Results	Explain	Describe	Saved SQL	History			
CLUBID	CLUBNAME	Total Competition Point	Total Goal Score	Home Win	Home Loss	Away Win	Away Loss
CL06	Wonnerup Wombats	38	20	3	3	4	1
CL01	Eagle Bay Eagles	37	22	5	2	3	2
CL08	Hanes Hawkes	36	17	2	1	4	2
CL02	Dunsborough Dingos	32	20	2	3	4	3
CL03	Broadwater Bandicoots	29	17	1	4	3	1
CL04	Mary-Brook Magpies	26	19	1	5	3	2
CL07	New Caves Rd Numbats	23	18	3	4	2	4
CL05	Reinscourt Rabbits	22	20	1	3	2	3

8 rows returned in 0.01 seconds [Download](#)

Explanation of query 4.1b

- The SELECT query had one main table which is Club (cl).

- The SQL query retrieves the clubID and clubName from the Club (cl) table.
- Subqueries were used to calculate additional information for each club.

From ClubGame(cg) and Result(rs) tables to SUM the competition points AS Total Competition Point.

From ClubGame(cg) to SUM the goals scored as Total Goal Score.

From ClubGame (cg), Result(rs), Game(gm) and Ground(grn) tables to count the number of Home win, Home loss, Away win and Away loss, matching for each club.

- Using aliases cl,cg,rs,gm,grn for the Club, ClubGame, Result, Game and Ground tables respectively. These make the query shorter and easier to read instead of full writing.

-WHERE condition inside each subquery tells that cg.clubID must match the current c.clubID. Thus, only the games related to each club are included.

-ORDER BY condition sorts the results by Total Competition Point in descending order. If the points are the same, then by Total Goal Score in descending order.

4.2 Seasonal Goals & Scorers

4.2a. Provide a list of the season's goals, who scored them and in which game. List the goals in chronological, then game order.

Syntax for query 4.2a

```
SELECT

gm.gameID AS "Game Number",

gm.description AS "Fixture",

cl.clubName AS "Club Scored",

pl.firstName || ' ' || pl.lastName AS "Name of the Scorer",

sg.gameMinute AS "Minute Goal Scored",

grn.name AS "Name of Ground",

gm.gameDate AS "Date"

FROM ScoredGoal sg

JOIN GamePlayer gp ON sg.gamePlayerID = gp.gamePlayerID

JOIN Player pl ON gp.playerID = pl.playerID

JOIN ClubGame cg ON gp.gameID = cg.gameID AND cg.clubID = pl.clubID

JOIN Club cl ON cg.clubID = cl.clubID

JOIN Game gm ON gp.gameID = gm.gameID

JOIN Ground grn ON gm.groundID = grn.groundID

ORDER BY gm.gameDate, gm.gameID, sg.gameMinute;
```

Screen-grab/Result of query 4.2a

Results	Explain	Describe	Saved SQL	History
---------	---------	----------	-----------	---------

Game Number	Fixture	Club Scored	Name of the Scorer	Minute Goal Scored	Name of Ground	Date
GM01	CL01 v CL02	Dunsborough Dingos	Steve Connor	40	Eagle Bay Centre	05/12/2025
GM01	CL01 v CL02	Eagle Bay Eagles	Glen Evans	44	Eagle Bay Centre	05/12/2025
GM01	CL01 v CL02	Eagle Bay Eagles	Glen Evans	66	Eagle Bay Centre	05/12/2025
GM01	CL01 v CL02	Eagle Bay Eagles	Glen Evans	78	Eagle Bay Centre	05/12/2025
GM01	CL01 v CL02	Eagle Bay Eagles	Kevin Murray	89	Eagle Bay Centre	05/12/2025
GM02	CL03 v CL04	Mary-Brook Magpies	Martin Tan	39	Broadwater Bay Oval	05/14/2025
GM02	CL03 v CL04	Broadwater Bandicoots	Frank Brooks	76	Broadwater Bay Oval	05/14/2025
GM03	CL05 v CL06	Wonnerup Wombats	Frank Hughes	50	West Busselton Oval	05/16/2025
GM03	CL05 v CL06	Wonnerup Wombats	Frank Hughes	76	West Busselton Oval	05/16/2025
...						
GM52	CL06 v CL04	Wonnerup Wombats	Frank Hughes	30	Wonnerup Sports Ground	08/10/2025
GM52	CL06 v CL04	Mary-Brook Magpies	Peter Connor	74	Wonnerup Sports Ground	08/10/2025
GM52	CL06 v CL04	Mary-Brook Magpies	Frank Brooks	79	Wonnerup Sports Ground	08/10/2025
GM52	CL06 v CL04	Wonnerup Wombats	David Brooks	84	Wonnerup Sports Ground	08/10/2025
GM52	CL06 v CL04	Wonnerup Wombats	Victor Johnson	89	Wonnerup Sports Ground	08/10/2025
GM54	CL03 v CL02	New Caves Rd Numbats	Glen Smith	41	Broadwater Bay Oval	08/13/2025
GM54	CL03 v CL02	New Caves Rd Numbats	Alex Martin	89	Broadwater Bay Oval	08/13/2025
GM55	CL07 v CL04	Broadwater Bandicoots	Peter Carter	65	Caves Road Reserve	08/15/2025
GM56	CL06 v CL05	Mary-Brook Magpies	Martin Tan	77	Wonnerup Sports Ground	08/17/2025
GM56	CL06 v CL05	Reinscourt Rabbits	Paul Brown	83	Wonnerup Sports Ground	08/17/2025

153 rows returned in 0.02 seconds [Download](#)

Explanation of query 4.2a

- The SELECT query has one main table which is Game (gm).
- The query shows details like the game number, fixture name, date, and ground name from the Game (gm) and Ground (grn) tables.
- Other tables are joined to add more information:
 ScoredGoal (sg) and GamePlayer (gp) are used to find which player scored the goal.
 Player (pl) is used to get the scorer's first and last name.
 ClubGame (cg) and Club (cl) are used to find which club the scorer belongs to.
 Aliases (gm, sg, gp, pl, cg, cl, grn) are used to make the query shorter and easier to read.
- The query also combines the player's first name and last name into one column so the full name is shown.
- The results are sorted by game date, game number, and the minute of the goal, so the goals are listed in the order they happened.

4.2b. Write a query that counts the number of goals scored at each of the 8 grounds during the season. Order the list you produce from most – to – least goals scored at the ground.

Syntax for query 4.2b

SELECT

grn.name AS "Ground Name",

```

COUNT(sg.goalID) AS "Total Goals",

ROUND(COUNT(sg.goalID) / COUNT(DISTINCT gm.gameID), 0) AS "Average Goals per Game"

FROM Ground grn

JOIN Game gm ON grn.groundID = gm.groundID

JOIN GamePlayer gp ON gm.gameID = gp.gameID

LEFT JOIN ScoredGoal sg ON gp.gamePlayerID = sg.gamePlayerID

GROUP BY grn.name;

```

Screen-grab/Result of query 4.2b

Ground Name	Total Goals	Average Goals per Game
Wonnerup Sports Ground	17	2
Kealy Reserve	22	3
West Busselton Oval	19	3
Broadwater Bay Oval	16	2
Caves Road Reserve	18	3
Hanes Sports Ground	14	2
Eagle Bay Centre	25	4
Quedjinup Reserve	22	3

8 rows returned in 0.01 seconds [Download](#)

Explanation of query 4.2b

- The main table in the query is Ground (grn), which gives us the ground name.
- The query counts how many goals (sg.goalID) were scored at each ground, shown as Total Goals.
- We also calculate the average number of goals per game at each ground by dividing: the total number of goals scored (COUNT of sg.goalID) by the number of different games played there (COUNT of DISTINCT gm.gameID).
- DISTINCT is used here because the same game could appear many times in the data (one row for each player). Using DISTINCT makes sure each game is only counted once.
- The JOINS connect the tables together:
 - Ground (grn) is joined to Game (gm) to find all games played at that ground.
 - Game (gm) is joined to GamePlayer (gp) to list the players in those games.
 - ScoredGoal (sg) is joined to GamePlayer (gp) to count which players actually scored goals.
- A LEFT JOIN is used so games with no goals are still included in the average.
- The GROUP BY grn.name ensures the totals and averages are calculated separately for each ground.

4.2c. Provide a list of the players who scored goals during the season, including the total number of goals each player scored. List this in descending order (i.e., number of goals scored) from highest number

Syntax for query 4.2c

```

SELECT

pl.firstname || ' ' || pl.lastname AS "Name",

```

```

COUNT(sg.gamePlayerID) AS "Goals Scored",
cl.clubname AS "Club Name"
FROM ScoredGoal sg
JOIN GamePlayer gp ON gp.gamePlayerID = sg.gamePlayerID
JOIN Player pl ON pl.playerID = gp.playerID
JOIN Club cl ON pl.clubID = cl.clubID
GROUP BY pl.firstname, pl.lastname, cl.clubname
ORDER BY COUNT(sg.gamePlayerID) DESC;

```

Screen-grab/Result of query 4.2c

Results	Explain	Describe	Saved SQL	History
Name	Goals Scored	Club Name		
Glen Evans	9	Eagle Bay Eagles		
David Martin	7	Dunsborough Dingos		
Frank Brooks	6	Broadwater Bandicoots		
Adam Tan	6	Dunsborough Dingos		
Alan Smith	6	Reinscourt Rabbits		
George Reid	6	Broadwater Bandicoots		
Frank Hughes	6	Wonnerup Wombats		
Paul Brown	6	Reinscourt Rabbits		
Kevin Murray	6	Eagle Bay Eagles		
Martin Tan	6	Mary-Brook Magpies		
Alex Martin	6	New Caves Rd Numbats		
Martin Connor	6	Hanes Hawkes		
Glen Smith	5	New Caves Rd Numbats		

...

Liam Foster	3	Mary-Brook Magpies
Jason Murray	2	Dunsborough Dingos
Kevin Hughes	2	Eagle Bay Eagles
Stuart Carter	2	New Caves Rd Numbats
Daniel Carter	2	Hanes Hawkes
Michael Wright	1	Broadwater Bandicoots
Daniel Wright	1	Dunsborough Dingos
Adam Hughes	1	Mary-Brook Magpies
Martin Wilson	1	Eagle Bay Eagles

36 rows returned in 0.00 seconds

[Download](#)

Explanation of query 4.2c

- The SELECT query has one main table which is Player (pl).
- The query shows details like the player's full name and last name together, the number of goals scored, and the club name from the Club table.
- Other tables are joined to add more information:
 - o ScoredGoal (sg) and GamePlayer (gp) are used to link each goal to the player who scored it.
 - o Player (pl) is used to display the player's first and last name.
 - o Club (cl) is used to show which club the player belongs to.
- Aliases (sg, gp, pl, cl) are used to make the query shorter and easier to read instead of writing the full table names.
- The query also uses COUNT(sg.gamePlayerID) to calculate how many goals each player has scored, and GROUP BY ensures each player only appears once with their total.
- The results are sorted by the total number of goals in descending order, so the players with the most goals are shown at the top of the list.

4.2d. Provide a list of the goals your highest goal scorer scored. What game the goal(s) took place in, against which club and on which ground. List the goals in chronological order.

Syntax for query 4.2d

SELECT

pl.firstName || ' ' || pl.lastName AS "Player Name",

gm.gameID AS "Game Number",

gm.description AS "Fixture",

opp.clubName AS "Scored Against",

grn.name AS "Ground Name",

sg.gameMinute AS "Minute Scored",

gm.gamedate AS "Game Date"

FROM ScoredGoal sg

JOIN GamePlayer gp ON sg.gamePlayerID = gp.gamePlayerID

JOIN Player pl ON gp.playerID = pl.playerID

JOIN Game gm ON gp.gameID = gm.gameID

JOIN ClubGame cg ON gp.gameID = cg.gameID AND cg.clubID = pl.clubID

```

JOIN ClubGame oppCG ON oppCG.gameID = gm.gameID AND oppCG.clubID <> cg.clubID

JOIN Club opp ON opp.clubID = oppCG.clubID

JOIN Ground grn ON gm.groundID = grn.groundID

WHERE gp.playerID IN (

SELECT gp2.playerID

FROM ScoredGoal sg2

JOIN GamePlayer gp2 ON sg2.gamePlayerID = gp2.gamePlayerID

GROUP BY gp2.playerID

HAVING COUNT(sg2.goalID) = (

SELECT MAX(totalGoals)

FROM (

SELECT gp3.playerID, COUNT(sg3.goalID) AS totalGoals

FROM ScoredGoal sg3

JOIN GamePlayer gp3 ON sg3.gamePlayerID = gp3.gamePlayerID

GROUP BY gp3.playerID

)

)

)

ORDER BY gm.gamedate, gm.gameID, sg.gameMinute;

```

Screen-grab/Result of query 4.2d

Player Name	Game Number	Fixture	Scored Against	Ground Name	Minute Scored	Game Date
Glen Evans	GM01	CL01 v CL02	Dunsborough Dingos	Eagle Bay Centre	44	05/12/2025
Glen Evans	GM01	CL01 v CL02	Dunsborough Dingos	Eagle Bay Centre	66	05/12/2025
Glen Evans	GM01	CL01 v CL02	Dunsborough Dingos	Eagle Bay Centre	78	05/12/2025
Glen Evans	GM13	CL01 v CL05	Reinscourt Rabbits	Eagle Bay Centre	69	06/02/2025
Glen Evans	GM21	CL01 v CL07	New Caves Rd Numbats	Eagle Bay Centre	81	06/16/2025
Glen Evans	GM25	CL01 v CL08	Hanes Hawkes	Eagle Bay Centre	81	06/23/2025
Glen Evans	GM29	CL02 v CL01	Dunsborough Dingos	Quedjinup Reserve	85	06/30/2025
Glen Evans	GM37	CL04 v CL01	Mary-Brook Magpies	Kealy Reserve	36	07/15/2025
Glen Evans	GM37	CL04 v CL01	Mary-Brook Magpies	Kealy Reserve	52	07/15/2025

9 rows returned in 0.00 seconds

[Download](#)

Explanation of query 4.2d

- The SELECT query has one main table which is ScoredGoal (sg).

- The query shows details such as the player name, game number, fixture description, opponent club, ground name, minute scored, and game date from the related tables.
- Other tables are joined to add more information:
 - GamePlayer (gp) is joined to connect each scored goal to the player who scored it.
 - Player (pl) is used to display the player's first name and last name together as the full name.
 - Game (gm) is used to show details of the match, such as the game number, description, and date.
 - ClubGame (cg) is joined with Player (pl) to identify the club that the scoring player belongs to.
 - ClubGame (oppCG) and Club (opp) are used again to find the opponent club in that game.
 - Ground (grn) is used to show the name of the ground where the game was played.
- This query first counts how many goals each player scored during the season (using the logic of the PlayerGoals table in the original version).
- Then, it finds the player or players who scored the most goals (the TopScorer step), by comparing each player's total goals to the maximum number of goals scored.
- Finally, the main query lists all the goals scored by these top scorer(s), showing the player's name, which game the goal was in, the fixture, the club they scored against, the ground where it happened, the minute of the goal, and the date of the game.
- Aliases (sg, gp, pl, gm, cg, cl, grn) are used to make the query shorter and easier to read instead of repeating full table names.
- The results are ordered by game date, game number, and minute scored, so the goals appear in the exact order they happened.

4.3 MVP (most valuable player) of the Season

4.3a. List all players who were given BoG-points during the season. Order the list from most BoG-points to Least BoG-points scored, and include their total number of BoG-points scored. This should tell you who won the Most Valuable Player in the South West Soccer Competition. You will score more SQL marks if you can somehow improve the result to show a summary of information such as how many times each player scored 3-points, or 2-points or 1-point as part of your whole presentation

Syntax for query 4.3a

```
SELECT

pl.playerID,

pl.firstName || ' ' || pl.lastName AS "Player Name",

/* Count how many times player scored 3 points */

(SELECT COUNT(*)

FROM BogPlayer bp3

JOIN BestOnGround bog3 ON bp3.bogID = bog3.bogID

JOIN GamePlayer gp3 ON bp3.gamePlayerID = gp3.gamePlayerID

WHERE gp3.playerID = pl.playerID AND bog3.points = 3) AS "Rank 1 (3 Points)",

/* Count how many times player scored 2 points */

(SELECT COUNT(*)

FROM BogPlayer bp2
```

```

JOIN BestOnGround bog2 ON bp2.bogID = bog2.bogID

JOIN GamePlayer gp2 ON bp2.gamePlayerID = gp2.gamePlayerID

WHERE gp2.playerID = pl.playerID AND bog2.points = 2) AS "Rank 2 (2 Points)",

/* Count how many times player scored 1 point */

(SELECT COUNT(*)

FROM BogPlayer bp1

JOIN BestOnGround bog1 ON bp1.bogID = bog1.bogID

JOIN GamePlayer gp1 ON bp1.gamePlayerID = gp1.gamePlayerID

WHERE gp1.playerID = pl.playerID AND bog1.points = 1) AS "Rank 3 (1 Point)",

/* Total BoG points */

(SELECT SUM(bog.points)

FROM BogPlayer bp

JOIN BestOnGround bog ON bp.bogID = bog.bogID

JOIN GamePlayer gp ON bp.gamePlayerID = gp.gamePlayerID

WHERE gp.playerID = pl.playerID) AS "Total BoG Points"

FROM Player pl

WHERE pl.playerID IN (

SELECT DISTINCT gp.playerID

FROM BogPlayer bp

JOIN GamePlayer gp ON bp.gamePlayerID = gp.gamePlayerID

)

ORDER BY "Total BoG Points" DESC;

```

Screen-grab/Result of query 4.3a

Results

Explain

Describe

Saved SQL

History

PLAYERID	Player Name	Rank 1 (3 Points)	Rank 2 (2 Points)	Rank 3 (1 Point)	Total BoG Points
PL810	Chris Brooks	4	1	1	15
PL110	Glen Evans	2	3	2	14
PL711	Alex Martin	3	2	1	14
PL609	Frank Hughes	2	3	1	13

....

PL112	Robert Tan	0	0	1	1
PL606	Stuart Coleman	0	0	1	1
PL503	Robert Tan	0	0	1	1
PL107	Peter Murray	0	0	1	1

60 rows returned in 0.05 seconds

[Download](#)

Explanation of query 4.3a

- The main table in this query is Player (pl). From here, we take each player's ID and combine their first name and last name into one column called Player Name.
- The query then uses small extra searches (subqueries) to calculate more details about each player:

We add together (SUM) all the Best-on-Ground (BoG) points that a player received across all games. This gives the Total BoG Points. It also counts how many times a player got 3 points, shown as Rank 1 (3 Points).

We count how many times a player got 2 points, shown as Rank 2 (2 Points).

We count how many times a player got 1 point, shown as Rank 3 (1 Point).

- Short names called aliases (pl, bp, bog, gp) are used so the query is easier to read instead of typing the full table names every time.
- The WHERE condition makes sure only players who actually received BoG points are included, so players with zero points don't show up.
- Finally, the results are sorted by Total BoG Points from highest to lowest. This means the player at the top of the list is the one with the most BoG points, which is the Most Valuable Player (MVP) of the season.

4.3b. Write a query that summarises your MVP's season. Include a way to display when 3-points, 2-points and/or 1-point were scored and in which games. List this in chronological/game order.

Syntax for query 4.3b

```
SELECT gm.gameID, gm.gameDate, pl.playerID, pl.firstName || ' ' || pl.lastName AS "Player Name", bog.points
AS "BoG Points"

FROM Player pl

JOIN GamePlayer gp ON pl.playerID = gp.playerID

JOIN BogPlayer bp ON gp.gamePlayerID = bp.gamePlayerID

JOIN BestOnGround bog ON bp.bogID = bog.bogID

JOIN Game gm ON gp.gameID = gm.gameID

WHERE pl.playerID = 'PL810' /* PL810: Chris Brooks, MVP from 4.3a */

ORDER BY gm.gameDate;
```

Screen-grab/Result of query 4.3b

Results	Explain	Describe	Saved SQL	History
GAMEID	GAMEDATE	PLAYERID	Player Name	BoG Points
GM08	05/25/2025	PL810	Chris Brooks	3
GM19	06/13/2025	PL810	Chris Brooks	1
GM22	06/18/2025	PL810	Chris Brooks	3
GM25	06/23/2025	PL810	Chris Brooks	2
GM50	08/06/2025	PL810	Chris Brooks	3
GM53	08/11/2025	PL810	Chris Brooks	3

6 rows returned in 0.00 seconds [Download](#)

Explanation of query 4.3b

- The SELECT query had one main table which is Player (pl).
- The SQL query retrieves the gameID, gameDate, playerID, firstName and lastName and Bog points from the related tables.
- JOIN between GamePlayer (gp) and BogPlayer (bp) links the player's game record with BoG entry.
- JOIN between BogPlayer (bp) and BestOnGround(bog) retrieves BoG points.
- JOIN between Game(gm) and GamePlayer (gp) links each BoG entry with its corresponding game information(gameID, gameDate).
- WHERE condition specifies a particular playerID (PL810) which corresponds to MVP identified in 4.3a. This ensures that only the season data of MVP is shown.
- ORDER BY condition sorts the results by gameDate in ascending order. Thus, Bog points are shown across the season chronologically.

4.4 Miscellaneous fun stuff

4.4a Which club had the best "away" record (i.e., scored the most points, won and/or drew the most often when playing away from their home ground) and what was the record.

Syntax for query 4.4a

```
SELECT cl.clubID, cl.clubName,
/* SUM of Away point */
SUM(rs.points) AS "Total Away Points",
/* The count of Away Win */
(SELECT COUNT(*))
FROM ClubGame cg2
JOIN Result rs2 ON cg2.resultID = rs2.resultID
JOIN Game gm2 ON cg2.gameID = gm2.gameID
JOIN Ground grn2 ON gm2.groundID = grn2.groundID
WHERE cg2.clubID = cl.clubID
AND cg2.clubID != grn2.clubID
AND rs2.description = 'Away Win') AS "Away Wins",
```

/ The count of Away Draw */*

```
(SELECT COUNT(*)  
FROM ClubGame cg3  
JOIN Result rs3 ON cg3.resultID = rs3.resultID  
JOIN Game gm3 ON cg3.gameID = gm3.gameID  
JOIN Ground grn3 ON gm3.groundID = grn3.groundID  
WHERE cg3.clubID = cl.clubID  
AND cg3.clubID != grn3.clubID  
AND rs3.description = 'Away Draw') AS "Away Draws",
```

/ The count of Away Loss */*

```
(SELECT COUNT(*)  
FROM ClubGame cg4  
JOIN Result rs4 ON cg4.resultID = rs4.resultID  
JOIN Game gm4 ON cg4.gameID = gm4.gameID  
JOIN Ground grn4 ON gm4.groundID = grn4.groundID  
WHERE cg4.clubID = cl.clubID  
AND cg4.clubID != grn4.clubID  
AND rs4.description = 'Away Loss') AS "Away Losses"
```

```
FROM Club cl
```

```
JOIN ClubGame cg ON cl.clubID = cg.clubID  
JOIN Result rs ON cg.resultID = rs.resultID  
JOIN Game gm ON cg.gameID = gm.gameID  
JOIN Ground grn ON gm.groundID = grn.groundID
```

```
WHERE cg.clubID != grn.clubID
```

```
GROUP BY cl.clubID, cl.clubName
```

/ Finding the highest away team score */*

```
HAVING SUM(rs.points) = (
```

```

SELECT MAX(highest_pts)

FROM (SELECT cg.clubID, SUM(rs.points) AS highest_pts

FROM ClubGame cg

JOIN Result rs ON cg.resultID = rs.resultID

JOIN Game gm ON cg.gameID = gm.gameID

JOIN Ground grn ON gm.groundID = grn.groundID

WHERE cg.clubID != grn.clubID

GROUP BY cg.clubID)

);

```

Screen-grab/Result of query 4.4a

CLUBID	CLUBNAME	Total Away Points	Away Wins	Away Draws	Away Losses
CL06	Wonnerup Wombats	31	4	2	1

1 rows returned in 0.02 seconds [Download](#)

Explanation of query 4.4a

The SELECT query had one main table which is Club (cl).

The SQL query retrieves the clubID and clubName from the Club (cl) table.

Subqueries were used to calculate additional information for each club:

From ClubGame(cg) and Result(rs) to SUM the total away points as Total Away Points.

From ClubGame(cg2), Result(rs2), Game(gm2), Ground(grn2) to COUNT the number of Away Wins as Away Wins.

From ClubGame(cg3), Result(rs3), Game(gm3), Ground(grn3) to COUNT the number of Away Draws as Away Draws.

From ClubGame(cg4), Result(rs4), Game(gm4), Ground(grn4) to COUNT the number of Away Losses as Away Losses.

Using aliases cl, cg, rs, gm, grn for the main query and cg2,cg3,cg4, rs2,rs3,rs4, gm2,gm3,gm4, and grn2,grn3,grn4 for the subqueries makes the query shorter and easier to read instead of full table names.

WHERE condition inside each subquery makes sure that only away games are counted by checking cg.clubID != grn.clubID.

The GROUP BY clause groups the result by clubID and clubName to calculate total away performance per club.

HAVING clause compares each club's SUM of away points with the maximum away points, calculated from a subquery (alias highest_pts). This ensures that only the club with the best away record are shown.

4.4b Which home-and-away game had the largest crowd watching the game.

Syntax for query 4.4b

```

SELECT

gm.description AS "Game Fixture",

gm.numOfSpectators AS "Number of Spectators",

cl.clubName AS "Home Club",

gr.name AS "Ground Name",

gr.address AS "Ground Address",

gm.gameDate AS "Game Date",

gm.startTime AS "Start Time",

gm.endTime AS "End Time"

FROM Game gm

JOIN Ground gr ON gm.groundID = gr.groundID

JOIN CompetitionWeek cw ON gm.weekID = cw.weekID

JOIN Club cl ON gr.clubID = cl.clubID

ORDER BY gm.numOfSpectators DESC

FETCH FIRST 1 ROWS ONLY;

```

Screen-grab/Result of query 4.4b

Bottom Splitter	in	Describe	Saved SQL	History
-----------------	----	----------	-----------	---------

Game Fixture	Number of Spectators	Home Club	Ground Name	Ground Address	Game Date	Start Time	End Time
CL03 v CL08	1980	Broadwater Bandicoots	Broadwater Bay Oval	Broadwater WA	06/13/2025	09/01/2025	09/01/2025

1 rows returned in 0.01 seconds [Download](#)

Explanation of query 4.4b

Using aliases **gm**, **gr**, **cw**, and **cl** for Game, Ground, CompetitionWeek, and Club respectively, the query joins Ground on **gm.groundID = gr.groundID**. This connects each game to its specific ground and establishes the home ground relationship. Then CompetitionWeek is joined on **gm.weekID = cw.weekID** to link games to their respective competition weeks, ensuring we're only looking at regular season home-and-away games. Finally, Club is joined on **gr.clubID = cl.clubID** to connect the ground to its home club, providing the home team information.

The query selects game details including the fixture description, spectator count, home club name, ground information, and game timing. The **ORDER BY gm.numOfSpectators DESC** command sorts all games in descending order based on the number of spectators, with the highest attendance first.

Finally, the **FETCH FIRST 1 ROWS ONLY** command retrieves only the top result, which represents the home-and-away game with the largest crowd watching the game.

4.4c List all the referees who awarded cards during the season and order them from being strictest to least.

Syntax for query 4.4c

```
SELECT re.firstName || ' ' || re.lastName AS "Referee Name",  
COUNT(ac.awardedCardID) AS "Total Cards Awarded" FROM AwardedCard ac  
JOIN GameReferee gr ON ac.gameRefereeID = gr.gameRefereeID  
JOIN Referee re ON gr.refereeID = re.refereeID  
GROUP BY re.firstName, re.lastName  
ORDER BY COUNT(ac.awardedCardID) DESC;
```

Screen-grab/Result of query 4.4c

Bottom Splitter

in

Describe

Saved SQL

History

Referee Name	Total Cards Awarded
Michael Jones	5
Sarah Wilson	3
James Patel	3
William Taylor	2

4 rows returned in 0.01 seconds

Download

Explanation of query 4.4c

Using aliases **ref**, **gr** and **ac** for **Referee**, **GameReferee** and **AwardedCard** respectively, joined **GameReferee** on

ac.gameRefereeID = gr.gameRefereeID. It joins AwardedCards with specific game and referee combination. Then Referee is joined on **ref.refereeID = gr.refereeID** to join game referee relationship to actual referee details.

The **GROUP BY** command creates separates groups for each referee and **COUNT** function counts the number of cards each referee awarded.

Finally, the result is ordered in descending order of count of awarded cards by each referee using **ORDER BY** command.

~ End of Assignment ~